

Where Deep Learning Losses Come From

Likelihood, cross-entropy, and regularization from probability

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ES 667 · Deep Learning · IIT Gandhinagar

Three familiar losses, one probabilistic origin

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Objective. Derive each term from an explicit likelihood or prior.

Two probabilistic assumptions determine the two halves of the objective.

We will define the predictive distribution and likelihood; use i.i.d. to move from one example to a dataset;
then introduce log-likelihood, NLL, the prior $p(\theta)$, and MAP.

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Probability primer

A point prediction summarizes a distribution

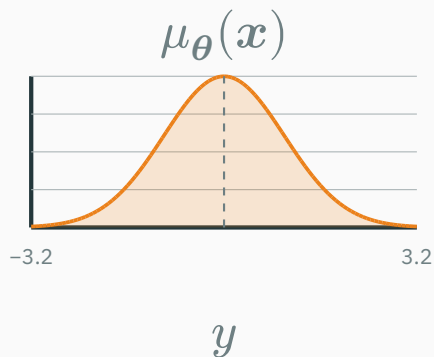
REGRESSION • $Y \in \mathbb{R}$

Point prediction: mean of $Y \mid \mathbf{X} = x$

$$\hat{y}(x) = \mathbb{E}_{\theta}[Y \mid \mathbf{X} = x] = \mu_{\theta}(x)$$

Full predictive distribution

$$Y \mid \mathbf{X} = x \\ \sim \mathcal{N}(\mu_{\theta}(x), \sigma_{\theta}^2(x))$$



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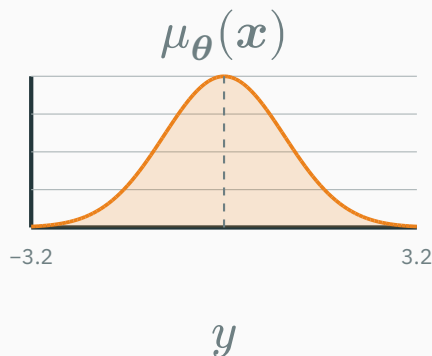
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BINARY CLASSIFICATION • $Y \in \{0, 1\}$

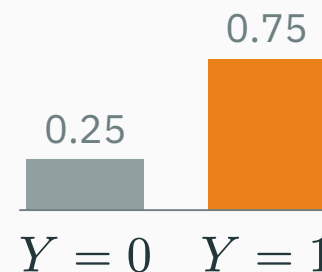
Point prediction: mode of $Y \mid \mathbf{X} = x$

$$\hat{y}(x) = \mathbf{1}[p_{\theta}(x) \geq 0.5]$$

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$$Y \mid \mathbf{X} = x \\ \sim \text{Bernoulli}(p_{\theta}(x))$$

example: $p_{\theta}(x) = 0.75$



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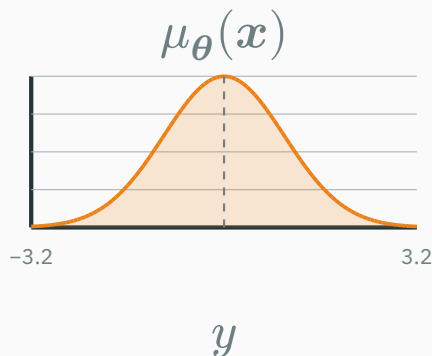
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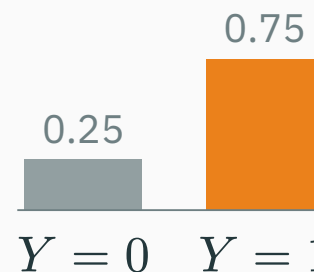
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Neural network: $x \rightarrow$ distribution parameters. **Prediction:** $\hat{y} =$ mean or mode.

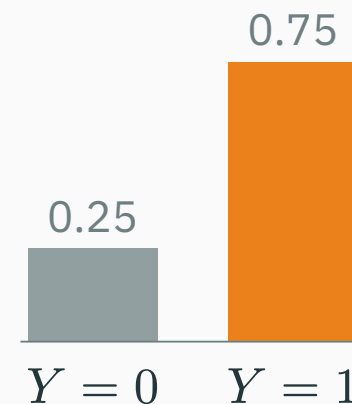
The **support** $\mathcal{S}_Y$ is the set of values that Y can take.

Probability mass function (PMF)

$$p_{Y(y)} = P(Y = y), \quad y \in \mathcal{S}_Y$$

$$p_{Y(y)} \geq 0, \quad \sum_{y \in \mathcal{S}_Y} p_{Y(y)} = 1$$

Bernoulli(0.75) PMF



each bar is an outcome probability

Examples. Bernoulli: $\{0, 1\}$ · categorical: $\{1, \dots, K\}$ · Poisson: $\{0, 1, 2, \dots\}$

Discrete distributions assign probability to outcomes

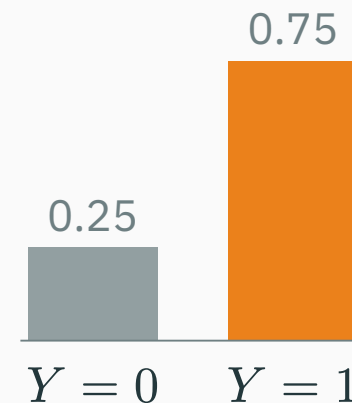
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Exact values can have nonzero probability: here $P(Y = 1) = 0.75$.

Continuous distributions assign density across a support

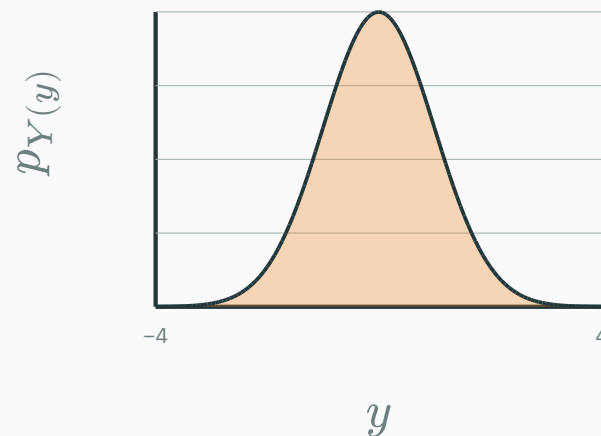
The **support** may be bounded or unbounded:

$$\mathcal{S}_Y = [a, b]$$

for a Uniform variable; $\mathcal{S}_Y = \mathbb{R}$ for a Gaussian.

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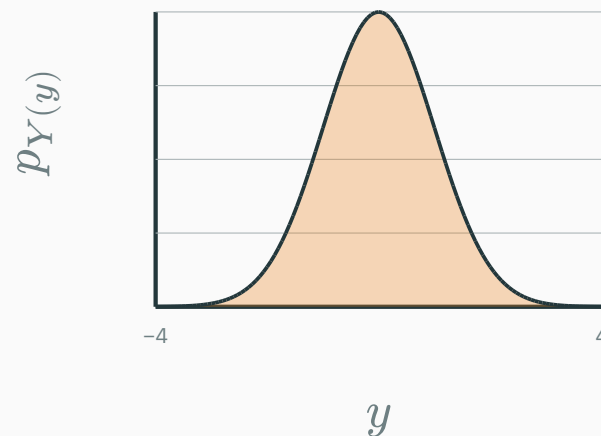
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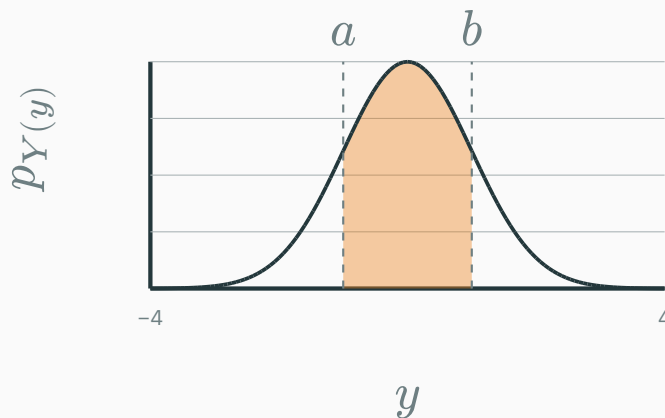
$$P(a \leq Y \leq b) = \int_a^b p_{Y(y)} dy$$

Density is not probability

For a continuous Y , probability comes from **area**:

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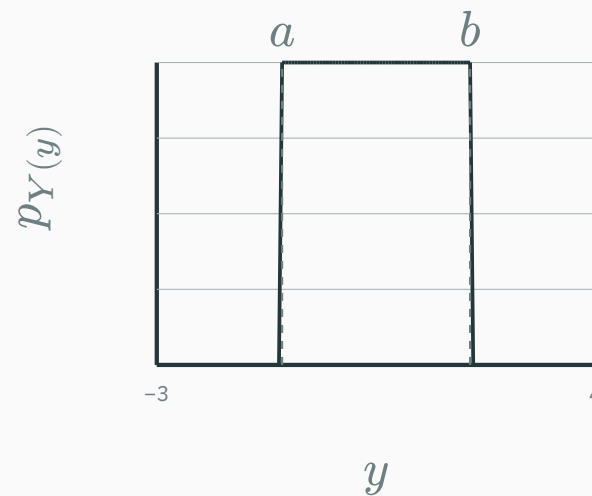
$p_{Y(y)} \neq P(Y = y)$ — a density can **exceed 1**; only *integrals* are probabilities.



the shaded area is $P(a \leq Y \leq b)$

$$Y \sim \mathcal{U}(a, b) \quad p_{Y(y)} = \begin{cases} 1/(b-a) & a \leq y \leq b \\ 0 & \text{otherwise} \end{cases}$$

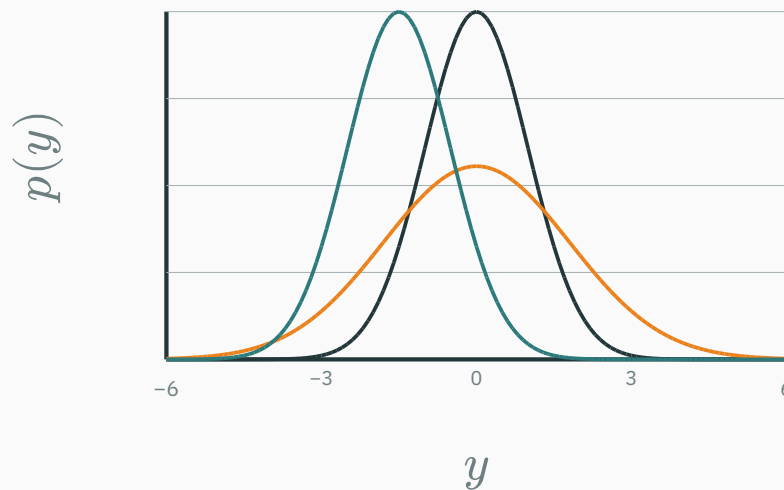
The support matters: outside $[a, b]$, the density is exactly 0.



$$Y \sim \mathcal{N}(\mu, \sigma^2) \quad p(y) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y - \mu)^2}{2\sigma^2}\right)$$

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— $\mu = 0, \sigma = 1$ — $\mu = 0, \sigma = 1.8$ — $\mu = -1.5, \sigma = 1$



Only two knobs: **location** μ and **scale** σ .

From probability to likelihood

Read the model notation one symbol at a time

Q QUESTION

$$p_{\theta}(y \mid x)$$

Read the model notation one symbol at a time

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X / x	X is the random input vector; x is one observed input
Y / y	Y is the random target; y is one possible or observed value
θ	the parameter vector — the model's learned weights and biases
$p_{\theta}(y \mid x)$	the probability (discrete Y) or density (continuous Y) assigned to y , given x

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Read aloud: “model θ 's probability or density for y , given x .”

Prediction fixes the model and varies the outcome

Example: $Y \in \{0, 1\}$. Fix the input x^* and one parameter setting θ_A .

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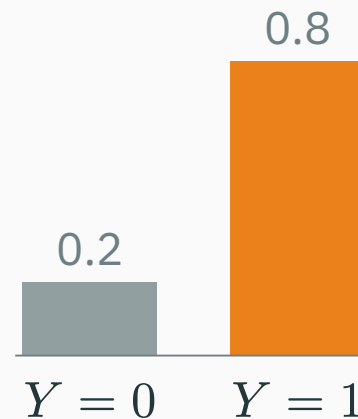
$$p_{\theta_A}(Y = 0 \mid x^*) = 0.20$$

$$p_{\theta_A}(Y = 1 \mid x^*) = 0.80$$

$$0.20 + 0.80 = 1$$

We vary the possible value of Y while keeping θ_A and x^* fixed.

predictive distribution



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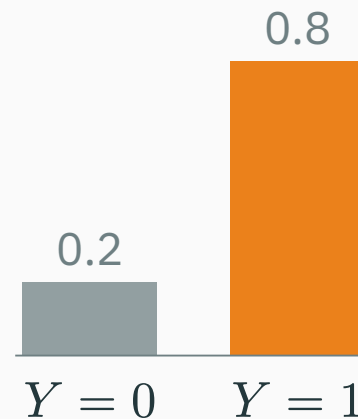
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$$\hat{y}(x^*) = \arg \max_y p_{\theta_A}(y \mid x^*) = 1$$

One observed label gives one likelihood score

D DERIVATION

Now observe $y^* = 1$ for the input x^* , so $\mathcal{D} = \{(x^*, 1)\}$.

$$L(\theta; \mathcal{D}) = p_{\theta}(y^* = 1 \mid x^*)$$

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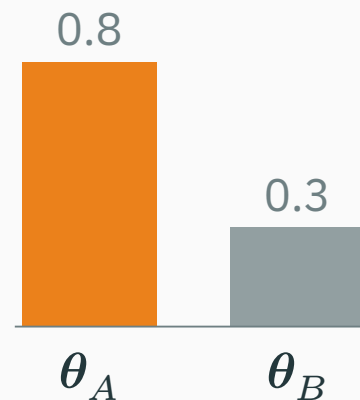
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θ_A makes the observed label $y^* = 1$ less surprising.

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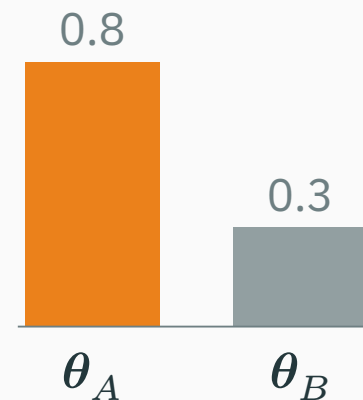
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Likelihood is a **score as a function of θ** ; the scores need not sum to 1 over parameter settings.

One continuous observation gives a likelihood too

Suppose $Y \mid \mu \sim \mathcal{N}(\mu, 1)$ and we observe the fixed value $y^* = 1$.

$$L(\mu; y^* = 1) = p(y^* = 1 \mid \mu) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(1 - \mu)^2}{2}\right).$$

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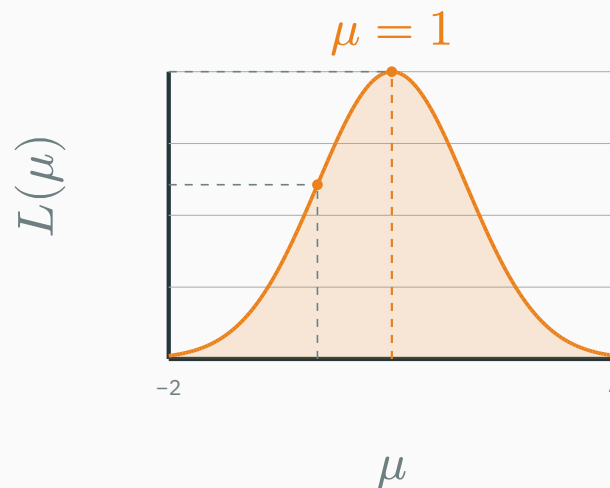
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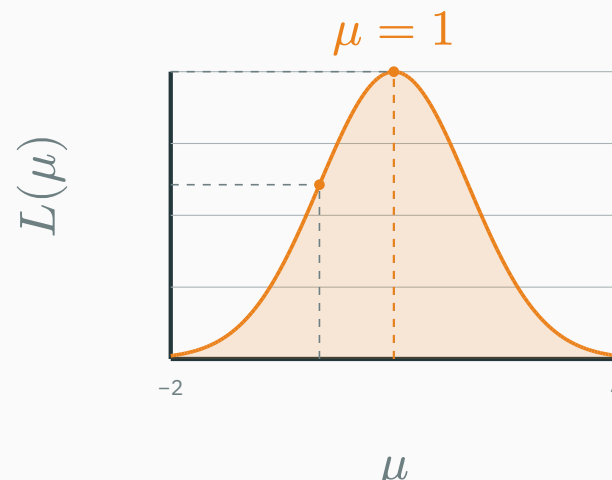
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Discrete or continuous: fix what was observed, then ask which parameter values explain it better.

A dataset needs a joint likelihood

For two observations, the likelihood starts as a **joint** probability or density:

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The product is a consequence of **conditional independence**; it is not automatic.

Independent means “no extra information”

D DERIVATION

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Coin-flip example. If $P(H \mid \theta) = \theta$ and the flips are independent, then $P(H, H \mid \theta) = \theta^2$.

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Independence is what licenses multiplication.

Identically distributed means “the same sampling rule”

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Coin flips

$Y_i \mid \theta \sim \text{Bernoulli}(\theta)$ for every i .
Each flip uses the same bias θ .

Supervised learning

The pairs $(\mathbf{X}_i, Y_i)$ come from the same population, and one shared θ defines $p_{\theta}(y_i \mid \mathbf{x}_i)$.

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i.i.d. = independent observations + one common data-generating rule.

Not independent: daily weather

Tomorrow's weather depends on today's. A first-order Markov model writes

$$p(y_{1:T} \mid \theta) = p(y_1 \mid \theta) \prod_{t=2}^T p(y_t \mid y_{t-1}, \theta).$$

The factors are **conditional on the previous day**, not independent marginals.

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Independent but not identical: two sensors

Two sensors measure the same temperature μ independently, but have different noise levels:

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i.i.d. is a useful special case—not a universal law.

Let R_1 mean “rain today” and R_2 mean “rain tomorrow”. Suppose

$$P(R_1) = 0.4, \quad P(R_2|R_1) = 0.8, \quad P(R_2|\neg R_1) = 0.2.$$

1. Use the conditional rule

$$\begin{aligned} P(R_1, R_2) &= P(R_1)P(R_2|R_1) \\ &= 0.4 \times 0.8 = 0.32. \end{aligned}$$

Both days are rainy with probability 0.32.

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Dependence does not forbid multiplication: use a conditional probability.

Given smoke S , two alarms act independently, but their sensitivities differ:

$$P(A_1 = 1|S) = 0.9, \quad P(A_2 = 1|S) = 0.6.$$

Independence

Both alarms ring with probability

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$$0.9 \neq 0.6$$

We cannot reuse one common factor for both alarms.

Worked example: independent does not mean identical

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Independent says “multiply”; identical says “reuse the same sampling rule”.

IID turns the joint likelihood into per-example factors

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$$p(\mathcal{D} \mid \theta) = \prod_{i=1}^n p_{\theta}(y_i \mid x_i).$$

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This is the bridge from one-example likelihood to a dataset loss.

Work two tiny Bernoulli datasets

D DERIVATION

Compare a fair coin $\theta = 0.5$ with a head-biased coin $\theta = 0.8$.

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Multiply one Bernoulli term per independent flip.

Two Gaussian measurements: multiply densities

D DERIVATION

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$$p(y_1 \mid \mu) \approx 0.352$$

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$$L(1.5) \approx 0.352 \times 0.352 = 0.124$$

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For continuous data we multiply **densities; $\mu = 1.5$ gives the larger likelihood.**

INTERACTIVE

↗ colab.research.google.com/github/nipunbatra/dl-teaching/blob/master/notebooks/L01/00_likelihood_iid_worked_examples.ipynb

Compute and plot single-observation likelihoods, multiply independent Bernoulli and Gaussian terms, compare candidate parameters, and verify that log-likelihood is a sum.

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In class: predict the winning parameter first; then run the cell and explain why the curve peaks there.

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Coin flips: a likelihood you can compute

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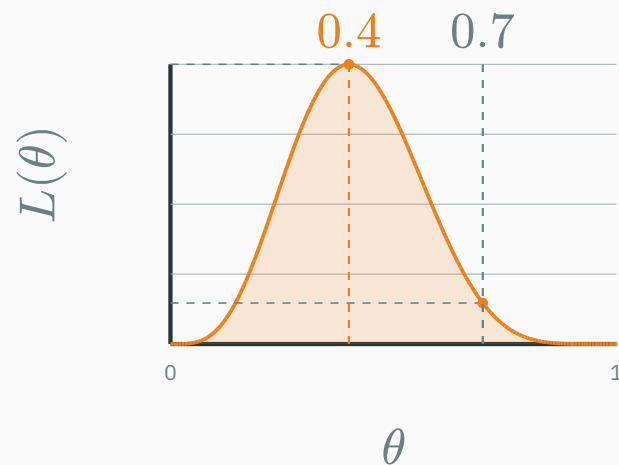
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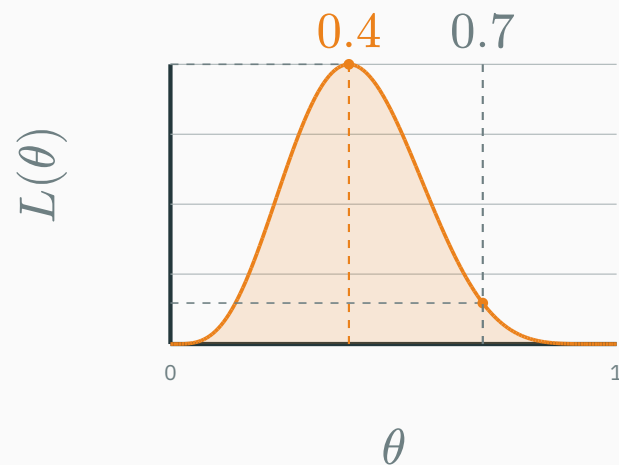
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The likelihood peaks at $\theta = 0.4$ — the observed head fraction $\frac{4}{10}$.

The **log-likelihood** is the logarithm of the likelihood — not a different model:

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The observed sequence has higher log-likelihood at $\theta = 0.4$.

Why logs? The maximizing parameter does not change

D DERIVATION

The logarithm is **strictly increasing** on positive numbers:

$$a > b > 0 \implies \log a > \log b$$

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Both likelihood and log-likelihood rank $\theta = 0.4$ above $\theta = 0.7$.

Poisson example: logging preserves the peak

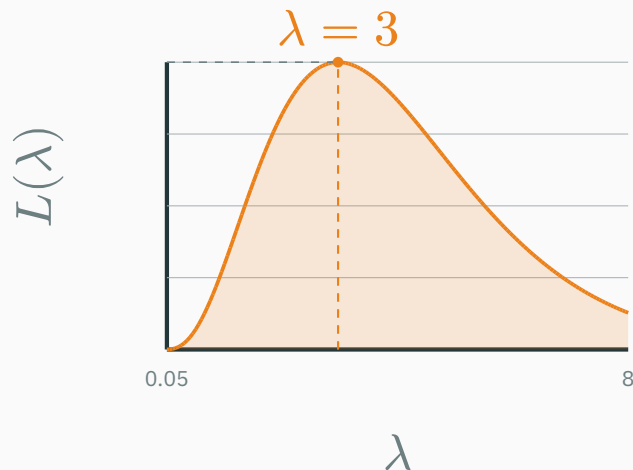
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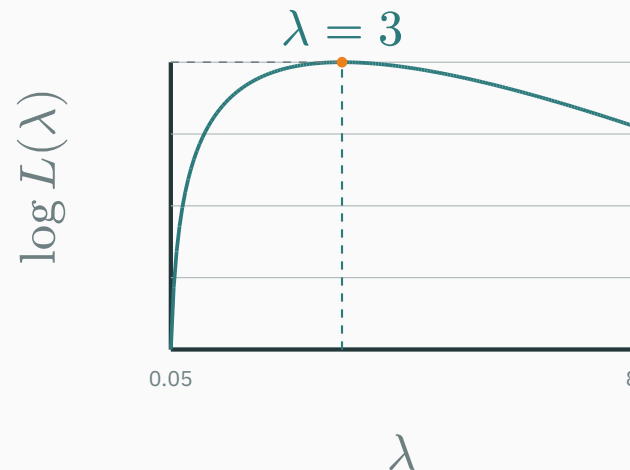
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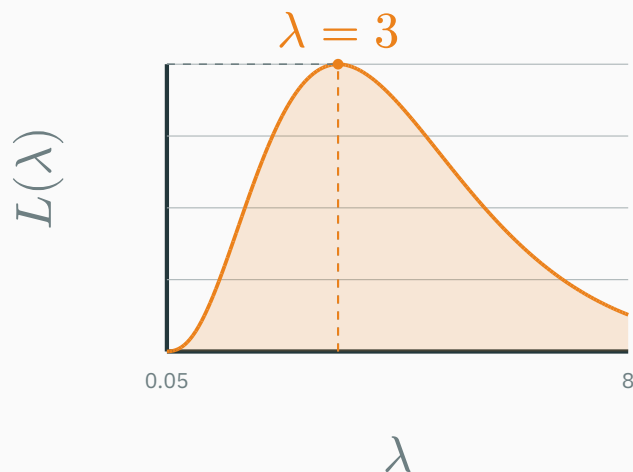


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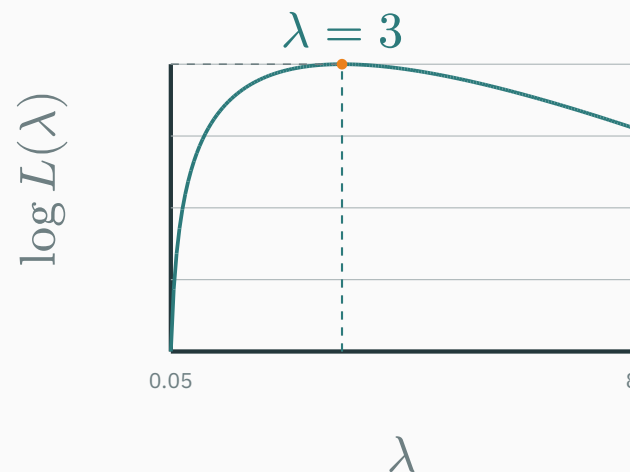
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Both curves peak at $\hat{\lambda}_{\text{MLE}} = 3$: the log changes the vertical scale, not the maximizing parameter.

Why logs? Products become stable sums

For independent observations,

$$L(\boldsymbol{\theta}) = \prod_i p_{\boldsymbol{\theta}}(y_i \mid \boldsymbol{x}_i) \implies \log L(\boldsymbol{\theta}) = \sum_i \log p_{\boldsymbol{\theta}}(y_i \mid \boldsymbol{x}_i)$$

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Worked numerical example. Suppose 200 observed outcomes each receive probability 0.01.

Direct product	$L = 0.01^{200} = 10^{-400} < 4.94 \times 10^{-324} \rightarrow \text{underflows to } 0$
Log space	$\log L = 200 \log(0.01) \approx -921.03 \rightarrow \text{finite and representable}$

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Logs make optimization easier and prevent numerical underflow.

Coin flips: solve for the stationary point

D DERIVATION

For $L(\theta) = \theta^{n_H} (1 - \theta)^{n_T}$, the log-likelihood is

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$\theta^* = 0.4$ is a **candidate** optimum. Next, check the curvature.

The second derivative confirms a maximum

Differentiate the log-likelihood again:

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At the stationary point $\theta^* = 0.4$,

$$-\frac{4}{(0.4)^2} - \frac{6}{(0.6)^2} = -25 - 16.67 \approx -41.67 < 0$$

Strict concavity + stationary point $\rightarrow$ unique global maximum: $\hat{\theta}_{\text{MLE}} = \theta^* = \frac{4}{10} = 0.4$.

Maximum likelihood estimation

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Interpretation. Choose the parameter values that assign the **highest likelihood** to the observed dataset.

Maximizing likelihood = minimizing NLL

D DERIVATION

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Per example	$\ell_{i(\boldsymbol{\theta})} = -\log p_{\boldsymbol{\theta}}(y_i \mid \mathbf{x}_i)$
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Minimizing empirical NLL is maximum likelihood; $\frac{1}{n}$ only rescales the objective.

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Zero likelihood becomes an infinite training penalty.

For a $\text{Uniform}(a, b)$ observation model, what is the NLL if the target lies outside $[a, b]$?

A. 0

B. A fixed finite penalty

C. $-\log 0 = +\infty$

D. It depends only on σ

Correct: C — $-\log 0 = +\infty$

Why: The model assigns zero density outside its support, so the observation is impossible under the assumption.

Regression: why MSE?

Regression as signal plus noise

$$y_i = f_{\boldsymbol{\theta}}(\mathbf{x}_i) + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2)$$

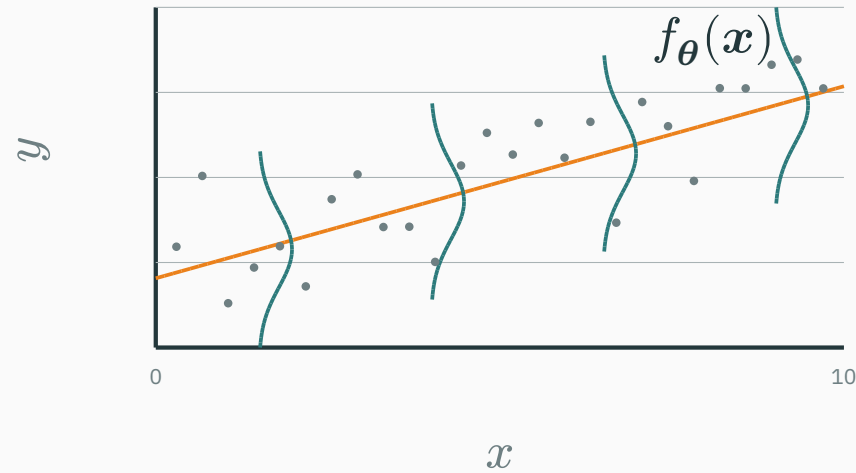
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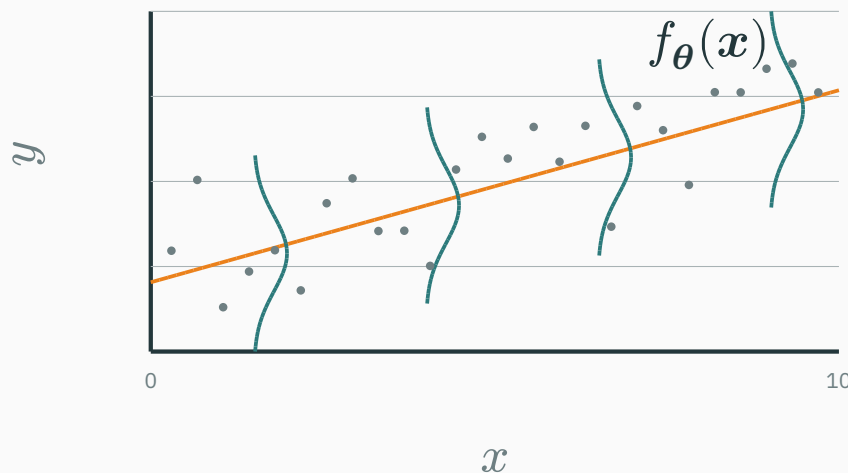
Therefore the conditional output distribution is

$$Y_i \mid \mathbf{X}_i = \mathbf{x}_i \sim \mathcal{N}(f_{\boldsymbol{\theta}}(\mathbf{x}_i), \sigma^2)$$

What the assumption means visually



What the assumption means visually



Gaussian noise lives in the **output direction**, conditional on x — a vertical bell at every input.

$$p_{\theta}(y_i \mid \mathbf{x}_i) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - f_{\theta}(\mathbf{x}_i))^2}{2\sigma^2}\right)$$

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Negative log of a single example:

$$-\log p_{\theta}(y_i \mid \mathbf{x}_i) = \frac{(y_i - f_{\theta}(\mathbf{x}_i))^2}{2\sigma^2} + C$$

MSE is Gaussian MLE

Sum over the dataset:

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MSE = Gaussian NLL, up to constants.

Linear regression makes f_θ explicit

D DERIVATION

For the one-dimensional picture,

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Linear regression makes f_{θ} explicit

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Then the same notation used throughout the lecture becomes

$$\hat{y}_i = f_{\theta}(x_i) = \theta^{\top} \tilde{x}_i$$

θ_0 is the intercept; $\theta_1, \dots, \theta_d$ are the slopes.

Substitute the linear mean into the observation model:

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With σ fixed, maximum likelihood therefore gives

$$\hat{\boldsymbol{\theta}}_{\text{MLE}} = \arg \min_{\boldsymbol{\theta}} \sum_i \underbrace{r_i^2}_{\text{squared vertical error}}$$

Ordinary least squares is Gaussian maximum likelihood for a linear mean.

The normal equations solve linear regression

D DERIVATION

Stack the augmented inputs into $\tilde{X} = [1 \quad X]$. Then

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If $\tilde{X}^{\top} \tilde{X}$ is invertible,

$$\hat{\theta} = (\tilde{X}^{\top} \tilde{X})^{-1} \tilde{X}^{\top} \mathbf{y}$$

Linear regression has a closed form; neural networks keep the MSE objective but use iterative optimization.

Why earlier ML courses could ignore σ^2

σ^2 is the observation-noise variance: the expected squared spread of targets around the mean prediction.

For n Gaussian observations, write $r_i = y_i - \mu_i$. The dataset NLL is

$$\mathcal{L}(\boldsymbol{\theta}; \sigma^2) = \frac{1}{2\sigma^2} \sum_i r_i^2 + \frac{n}{2} \log \sigma^2 + C.$$

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For point predictions, fixed-variance Gaussian MLE gives the same fitted model as MSE.

One global noise level.

Fit the mean model, then estimate the residual spread:

$$\hat{\sigma}_{\text{MLE}}^2 = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{\mu}_i)^2.$$

Use this when a residual plot has roughly the same vertical width across inputs.

How is σ^2 chosen in practice?

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Input-dependent noise.

Let the network predict both

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Use this when residual spread changes with the input—for example, a funnel shape.

Fit both cases with Gaussian NLL; validate whether uncertainty intervals attain their promised coverage.

HOMOSCEDASTIC



Network output: μ_i

Shared variance: σ^2

same uncertainty at every x_i

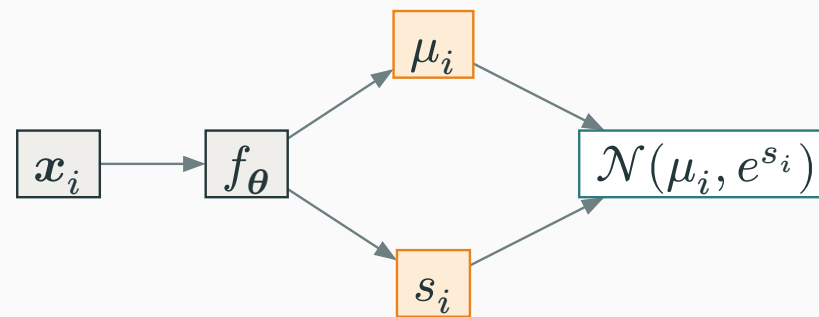
Homoscedastic vs input-dependent uncertainty

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HETEROSCEDASTIC



Network outputs: μ_i and s_i
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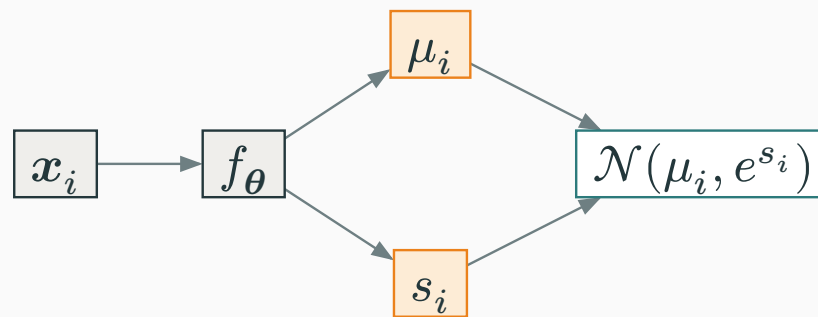
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Predict **log-variance** so every real s_i maps to a positive variance e^{s_i} .

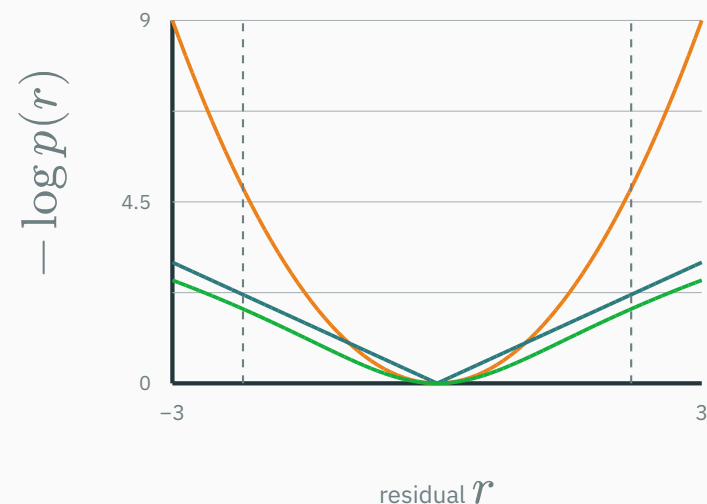
The noise model determines how errors are penalized

Noise model	Negative log-likelihood
— Gaussian	r^2
— Laplace	$ r $
— Uniform	0 inside; $+\infty$ outside
— Student- t	$\log\left(1 + \frac{r^2}{\nu}\right)$

$r = y - \hat{y}$ · swatches match plot

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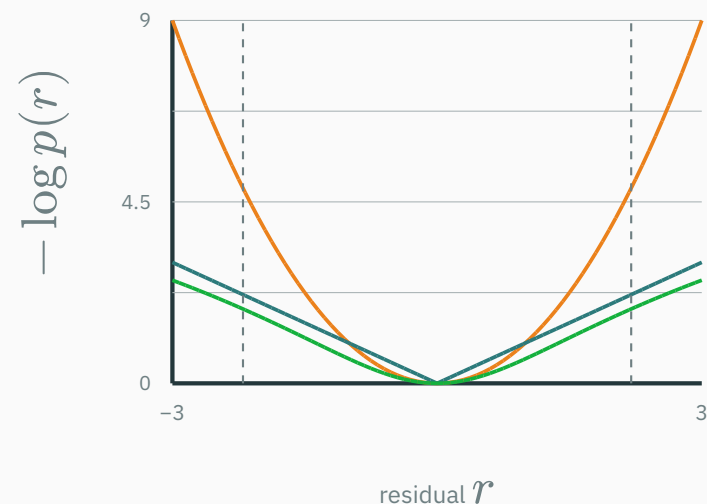


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Gaussian penalizes large errors most; Student- t is more robust. The dashed Uniform boundaries mark an infinite penalty outside the allowed interval.

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/likelihood-to-loss

likelihood-to-loss.html — three synchronized panels: the **noise density** $p(r)$, the **loss** $-\log p(r)$, and **data with a fitted line**.

Controls: Gaussian / Laplace / Uniform / Student- t · noise scale · slope & intercept · **add an outlier** · per-point contributions.

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Ask: which model reacts most to an outlier? why does Uniform give an infinite barrier? which loss is robust?

Classification: cross-entropy

Classification should output probabilities

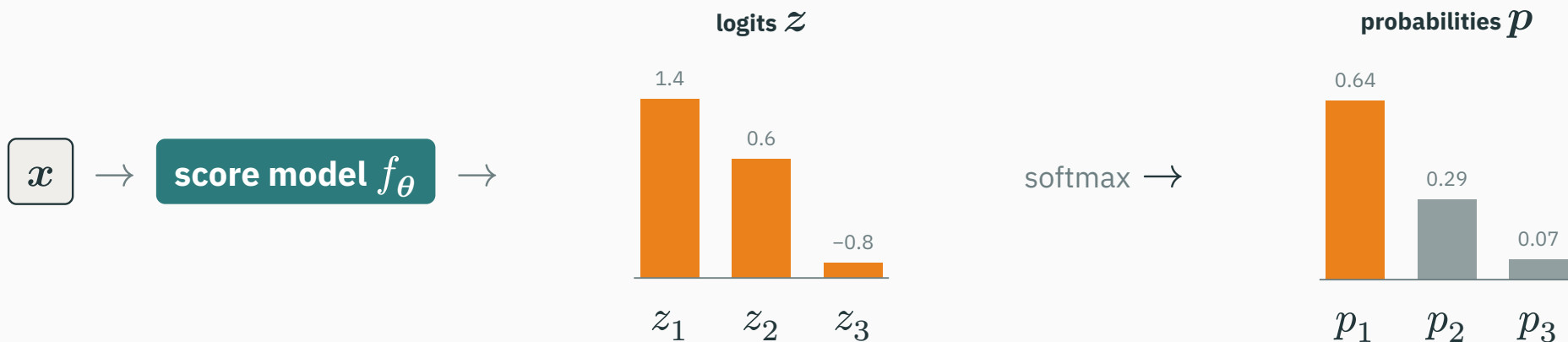
For K classes:

$$p_{\theta}(Y = k \mid \mathbf{x}) \geq 0, \quad \sum_{k=1}^K p_{\theta}(Y = k \mid \mathbf{x}) = 1$$

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f_{θ} may be **linear** (softmax regression) or **nonlinear** (a neural network).
It produces **logits** $z_k \in \mathbb{R}$; softmax converts them to **probabilities** $p_k \in [0, 1]$.

Binary classification: Bernoulli model

$$Y \mid \mathbf{X} = \mathbf{x} \sim \text{Bernoulli}(p_{\boldsymbol{\theta}}(\mathbf{x}))$$

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$$y = 1 \Rightarrow p(y) = p$$

$$y = 0 \Rightarrow p(y) = 1 - p$$

$$-\log p_{\theta}(y \mid \boldsymbol{x}) = -\log(p^y(1-p)^{1-y})$$

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$$\begin{aligned} -\log p_{\theta}(y \mid \boldsymbol{x}) &= -\log(p^y(1-p)^{1-y}) \\ &= -y \log p - (1-y) \log(1-p) \end{aligned}$$

Binary cross-entropy = Bernoulli NLL.

Check. True label $y = 1$: if $p = 0.8$ the loss is $-\log 0.8 \approx 0.22$; if $p = 0.2$ it is $-\log 0.2 \approx 1.61$.

The network emits an unbounded score $z = f_{\theta}(x) \in \mathbb{R}$; the **sigmoid** maps it to a probability:

$$p = \sigma(z) = \frac{1}{1 + e^{-z}}$$

Why use logits?

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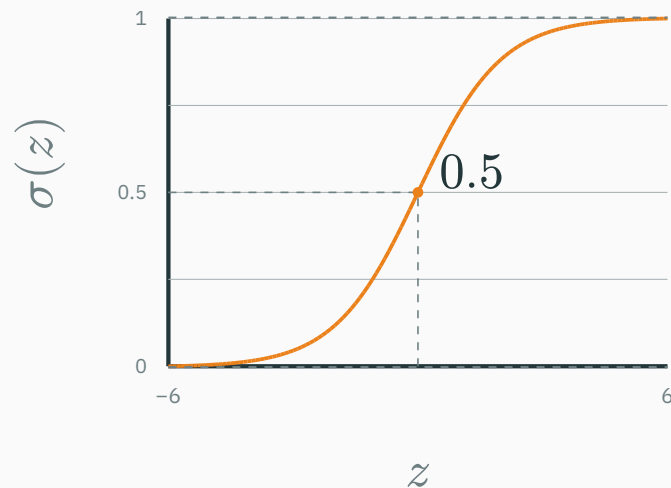
$$p = \sigma(z) = \frac{1}{1 + e^{-z}}$$

$$z \rightarrow -\infty : p \rightarrow 0$$

$$z = 0 : p = 0.5$$

$$z \rightarrow +\infty : p \rightarrow 1$$

In practice sigmoid + BCE are fused for numerical stability.



Use an augmented input $\tilde{x}_i$ so θ contains both weights and the intercept:

$$p_i = \sigma(\theta^\top \tilde{x}_i), \quad Y_i \mid \mathbf{X}_i = \mathbf{x}_i \sim \text{Bernoulli}(p_i)$$

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Logistic regression is Bernoulli maximum likelihood; its training objective is **binary cross-entropy**.

CATEGORICAL OBSERVATION

$$Y \mid \mathbf{X} = \mathbf{x} \sim \text{Categorical}(p_1, \dots, p_K)$$

A **one-hot** target has one 1 at the true class and 0 everywhere else.

Example: class 2 of 3 $\rightarrow \mathbf{y} = (0, 1, 0)$

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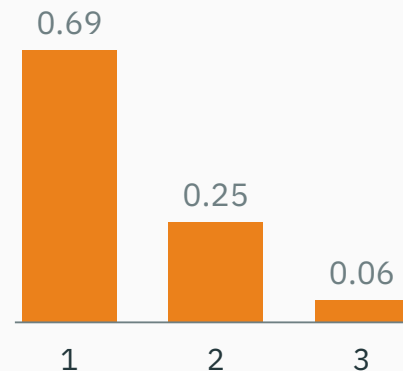
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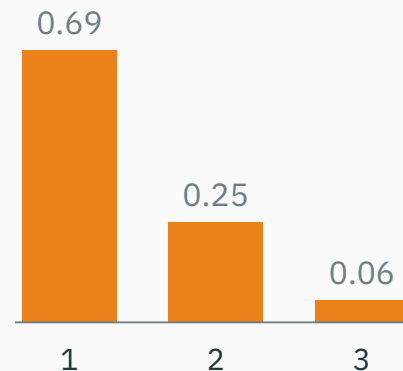
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Softmax guarantees $p_k > 0$ and $\sum_k p_k = 1$, as a categorical model requires.

Categorical NLL gives cross-entropy

D DERIVATION

$$-\log p_{\theta}(\mathbf{y} \mid \mathbf{x}) = -\log \prod_k p_k^{y_k} = -\sum_k y_k \log p_k$$

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Because $\mathbf{y}$ is one-hot, only the true class survives:

$$\mathcal{L} = -\log p_{\text{true class}}$$

$$\ell(p_y) = -\log p_y$$

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Read off the loss at three confidences for the **true** class:

Confident mistakes are expensive

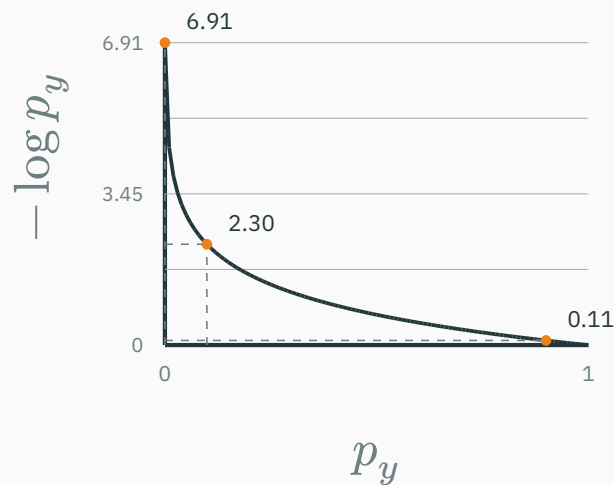
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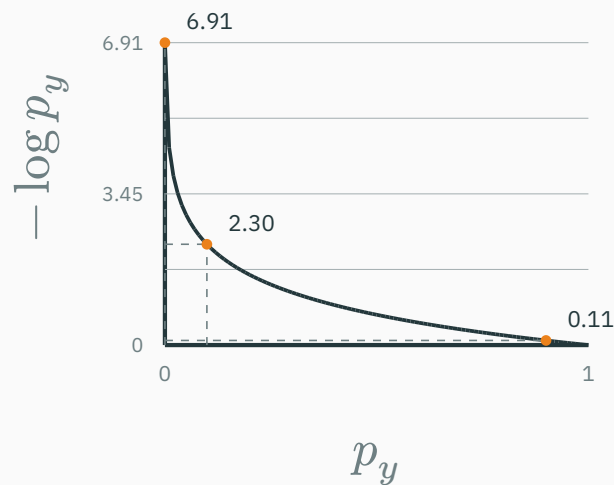
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The negative log makes a confident wrong prediction a large loss.

Why does cross-entropy penalize a confidently wrong prediction strongly?

A. It makes all class probabilities equal

B. The true-class probability is near zero, so $-\log p_y$ is large

C. It adds an L_2 penalty to the logits

D. The gradient is always zero

Correct: B — The true-class probability is near zero

Why: The negative log grows without bound as p_y approaches zero, matching the fact that the model declared the observed class nearly impossible.

A binary label takes only two values

★ OPTIONAL

For one observed example, let the target be $y = 1$ and the model predict

$$p = P(Y = 1|\mathbf{x}) = 0.8.$$

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Squared error

$$\ell_{\text{MSE}} = (y - p)^2 = (1 - 0.8)^2 = 0.04.$$

Its usual likelihood interpretation is
Gaussian:

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Permits all real Y —not just labels.

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Binary cross-entropy

$$\ell_{\text{CE}} = -\log p = -\log 0.8 \approx 0.223.$$

Its likelihood is Bernoulli:

$$Y|\mathbf{x} \sim \text{Bernoulli}(0.8).$$

Permits only $Y = 0$ or $Y = 1$.

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MSE is possible; cross-entropy is the negative log-likelihood matched to a binary observation.

Bayes' rule for learning

PROBABILITY LECTURE

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}$$

A — event or hypothesis to infer

B — observed evidence

Examples: disease given a test; urn given a sampled colour.

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Question: which parameter settings remain plausible after seeing the data?

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Replace A by $\boldsymbol{\theta}$ and B by $\mathcal{D}$: the algebra is unchanged; the interpretation changes.

Three heads do not prove $\theta = 1$

Let $\theta = P(H)$ and suppose the first three flips are $\mathcal{D} = (H, H, H)$.

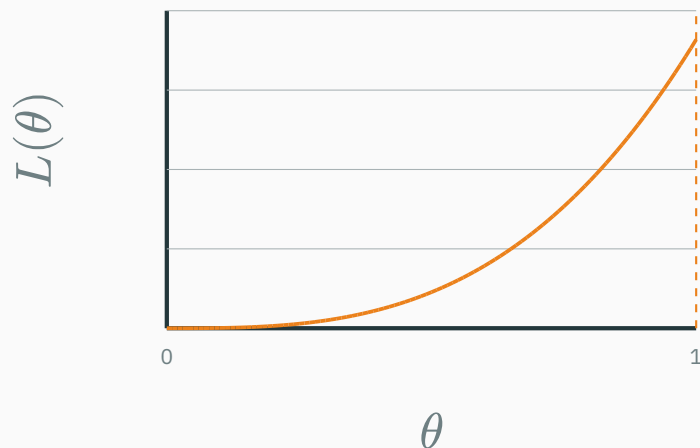
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H · H · H

$$L(\theta; \mathcal{D}) = p(\mathcal{D} \mid \theta) = \theta^3$$

$$L(\theta) = \theta^3$$

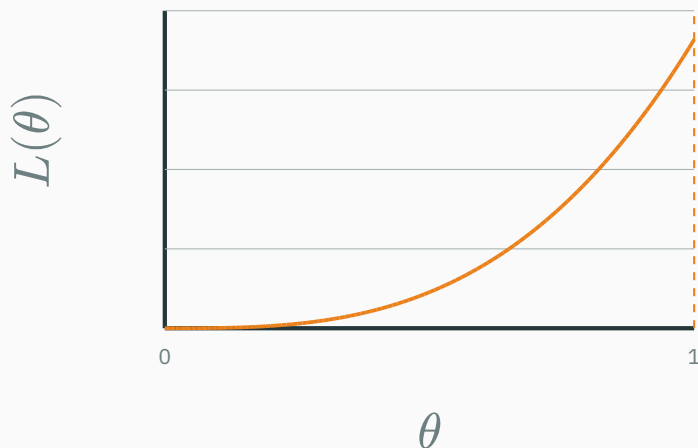


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H • H • H

$$L(\theta) = \theta^3$$



$$L(\theta; \mathcal{D}) = p(\mathcal{D} \mid \theta) = \theta^3$$

$$\hat{\theta}_{\text{MLE}} = \arg \max_{0 \leq \theta \leq 1} \theta^3 = 1$$

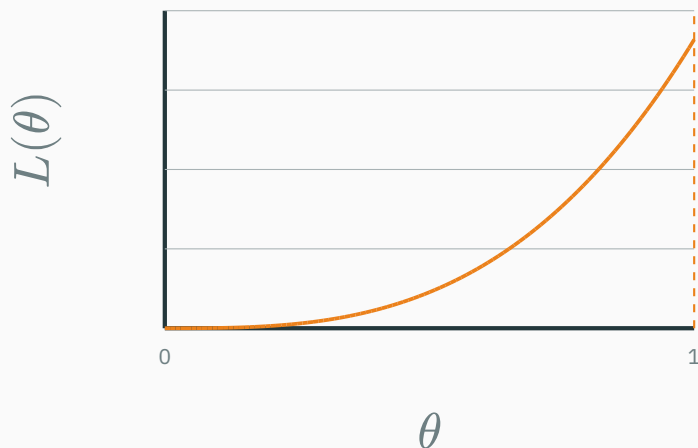
Did we learn that the coin **always** lands heads—or did a merely head-biased coin produce a lucky run?

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H • H • H

$$L(\theta) = \theta^3$$



$$L(\theta; \mathcal{D}) = p(\mathcal{D} \mid \theta) = \theta^3$$

$$\hat{\theta}_{\text{MLE}} = \arg \max_{0 \leq \theta \leq 1} \theta^3 = 1$$

Did we learn that the coin **always** lands heads—or did a merely head-biased coin produce a lucky run?

MLE identifies the best-fitting θ ; it does **not make $\theta = 1$ certain.**

A prior tempers the conclusion from three flips

Choose a smooth prior on $\theta \in [0, 1]$, highest near the fair-coin value 0.5.

Prior

$$p(\theta) = 6\theta(1 - \theta)$$

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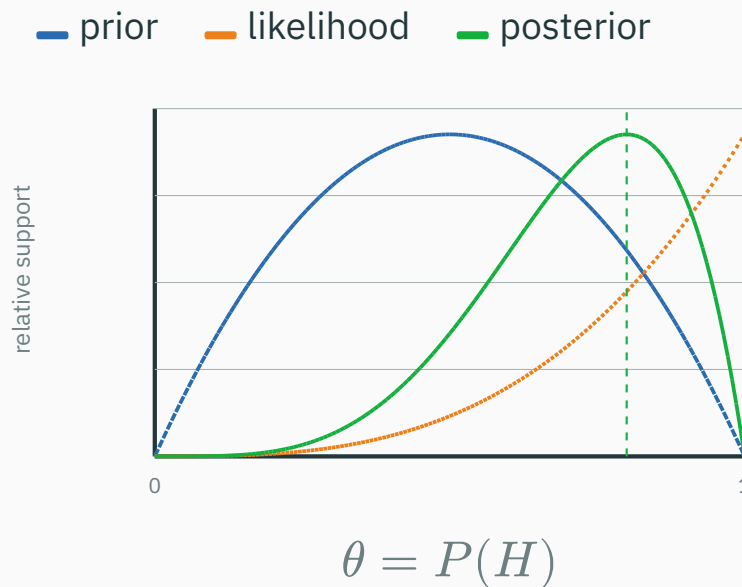
$$p(\mathcal{D} \mid \theta) = \theta^3$$

Posterior

$$p(\theta \mid \mathcal{D}) \propto \theta^3 p(\theta) \propto \theta^4 (1 - \theta)$$

Likelihood peak (MLE): $\theta = 1$

Posterior peak: $\theta = \frac{4}{5} = 0.8$



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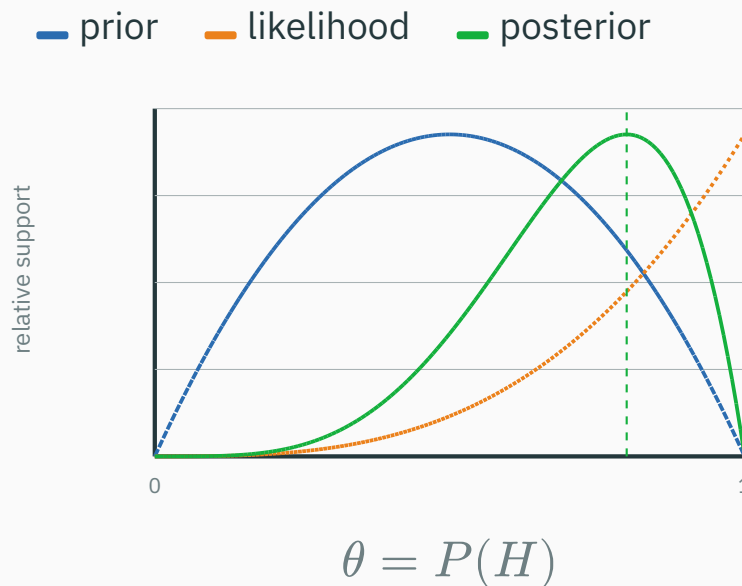
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With little data, the prior matters; as observations accumulate, the likelihood increasingly dominates.

Priors, MAP and regularization

Every learner has an inductive bias

When several predictors fit the training data, the learner still needs a preference for **unseen inputs**. That preference is its **inductive bias**.

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Model	Built-in preference → consequence
Linear regression	responses vary linearly with features → line or plane extrapolation
k-NN	nearby inputs tend to have similar outputs → local prediction
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A prior is one explicit inductive bias; the model class, features, architecture, and optimization add others.

The same evidence can start from different priors

D DERIVATION

Suppose only two coins are possible: **fair** $F : \theta = 0.5$ or **heads-biased** $B : \theta = 0.9$.

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Observe $\mathcal{D} = (H, H, H)$: $P(\mathcal{D}|F) = 0.5^3 = 0.125$ $P(\mathcal{D}|B) = 0.9^3 = 0.729$.

Prior A: probably fair

$$P(F) = 0.90, \quad P(B) = 0.10$$

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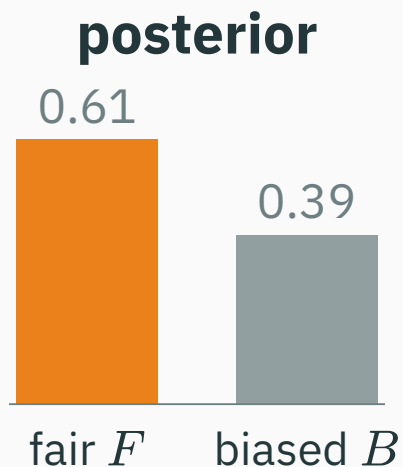
$$\begin{aligned} \text{prior} \times \text{likelihood } F : & \quad 0.10 \times 0.125 = 0.0125 \\ B : & \quad 0.90 \times 0.729 = 0.6561 \end{aligned}$$

Bayes' rule first forms a weight for each hypothesis: prior $\times$ likelihood.

Normalize the weights to obtain the posterior

Divide each weight by the sum of the two weights so the posterior probabilities add to 1.

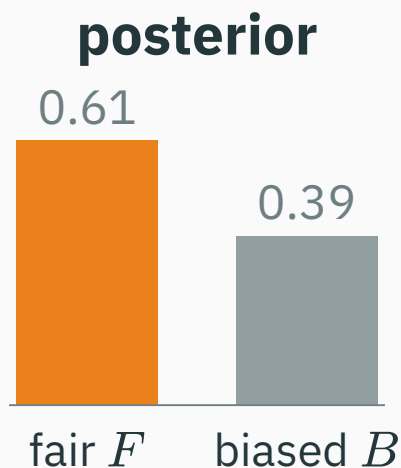
Prior A: probably fair



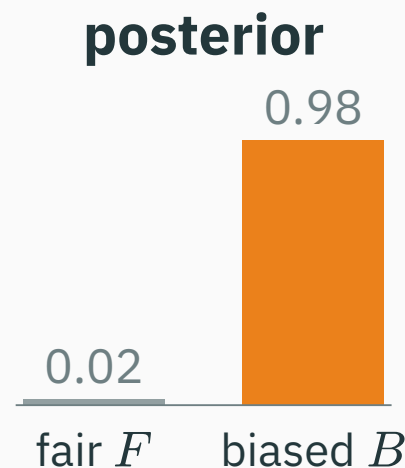
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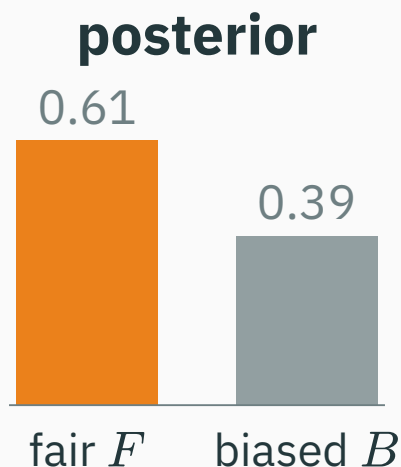
Prior B: probably biased



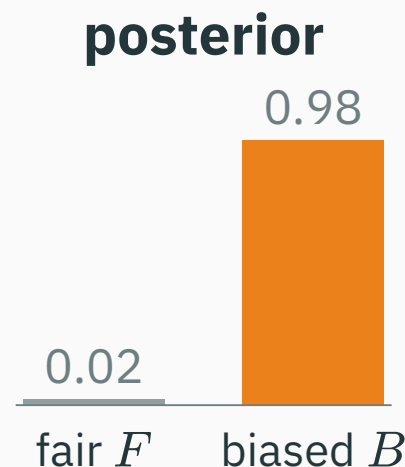
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Prior A: probably fair



Prior B: probably biased



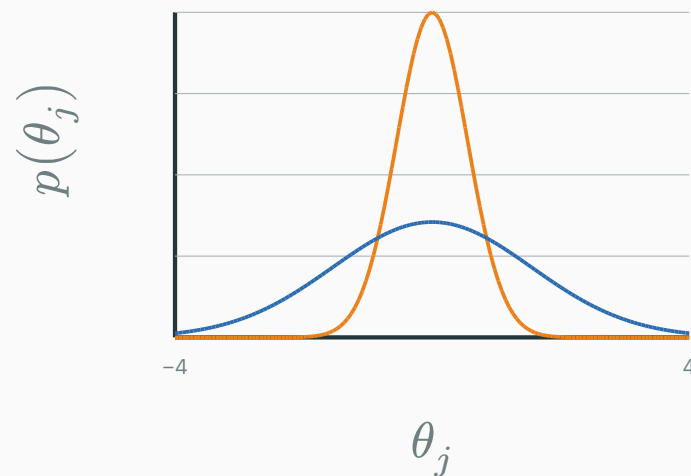
The observation is identical; with little data, different priors produce different posteriors.

$$\theta \sim p(\theta)$$

For a neural network, θ contains all weights and biases. Before using the current dataset, a prior states which settings are more plausible.

It can prefer small or sparse weights, smooth functions, and correlated parameters.

— narrow $\sigma = 0.55$ — broad $\sigma = 1.55$



narrow = stronger preference near 0

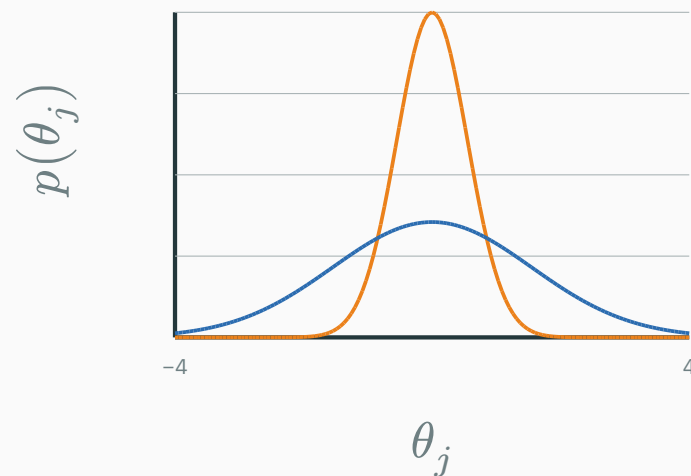
A neural-network prior ranks plausible parameter vectors

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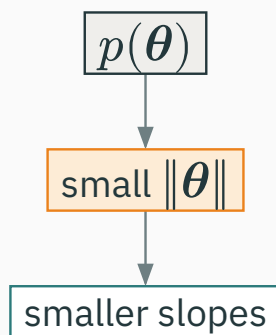


narrow = stronger preference near 0

A prior makes part of the model's inductive bias explicit as a probability distribution.

REGRESSION

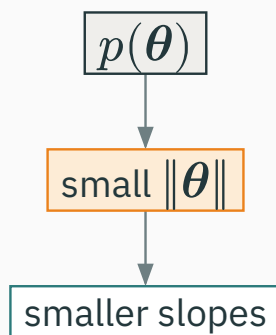
$$Y_i \mid \mathbf{X}_i = \mathbf{x}_i \sim \mathcal{N}(\boldsymbol{\theta}^\top \tilde{\mathbf{x}}_i, \sigma^2)$$



Before seeing data, flatter linear predictions are more plausible.

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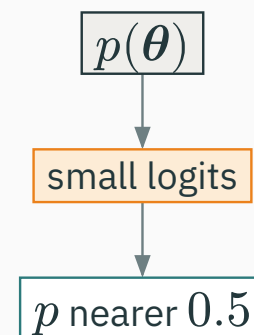
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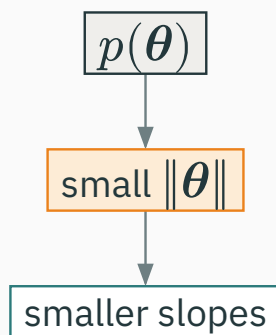
$$p_{\boldsymbol{\theta}}(Y = 1 \mid \mathbf{x}) = \sigma(\boldsymbol{\theta}^\top \tilde{\mathbf{x}})$$



Before seeing data, extremely confident predictions are less plausible.

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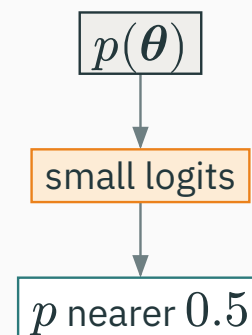
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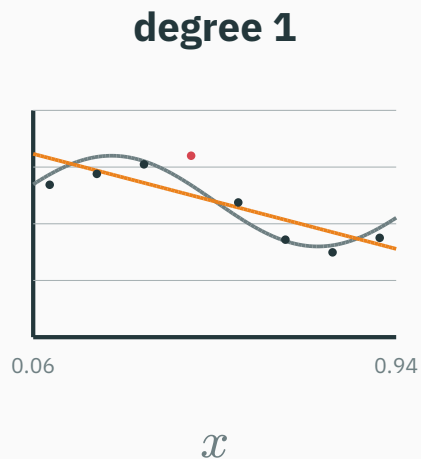
Before seeing data, extremely confident predictions are less plausible.

The prior acts on parameters; its visible effect depends on how those parameters enter the model.

Model complexity: underfitting to overfitting

Fit the same eight observations by least squares; only the polynomial degree changes.

dashed = underlying trend · orange = fitted polynomial · red dot = one noisy observation



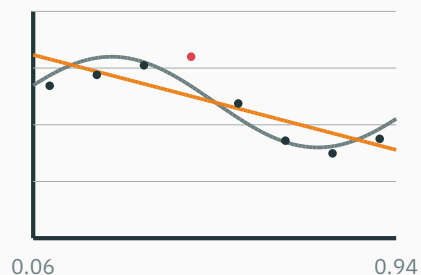
underfit

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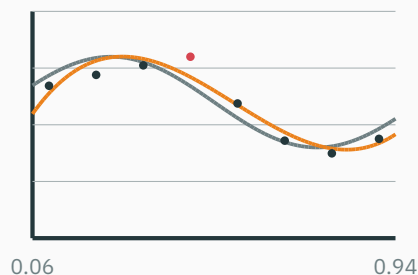
degree 1



$\mathcal{X}$

underfit

degree 3



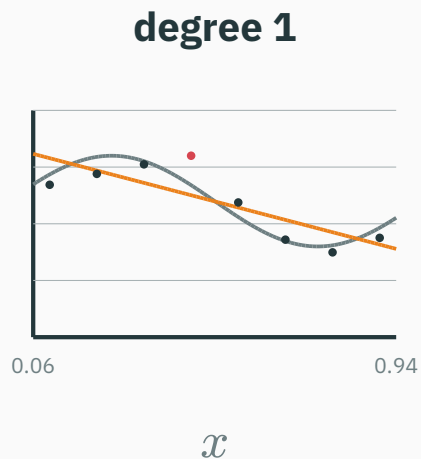
$\mathcal{X}$

captures the main trend

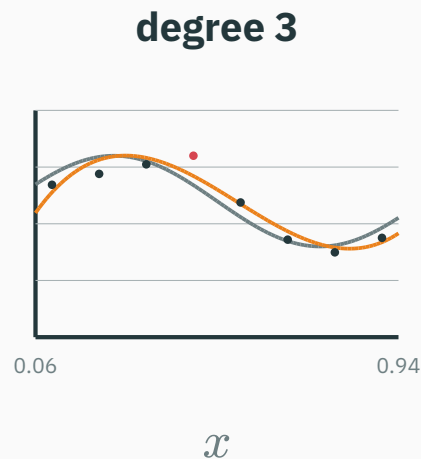
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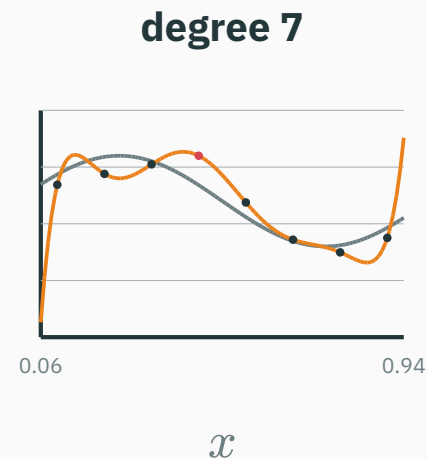
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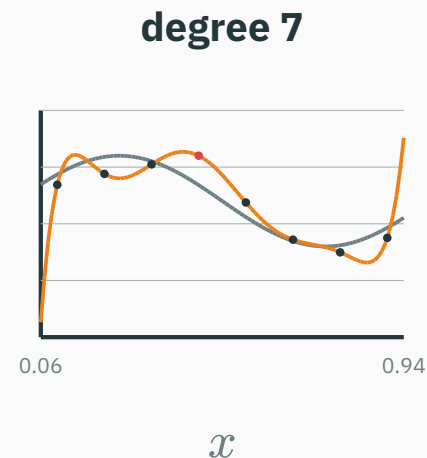
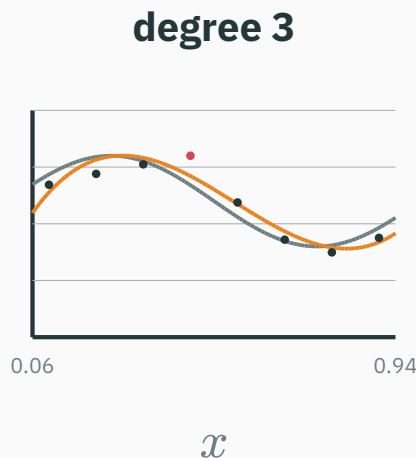
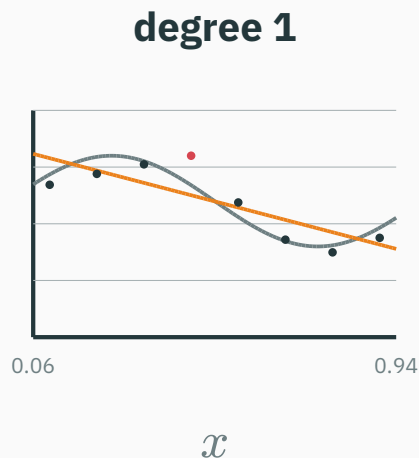


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underfit

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overfit

Higher degree lowers training error, but degree 7 chases the noisy observation and becomes unstable. Priors and regularization can prefer smoother solutions.

Start with four noisy points from a straight line

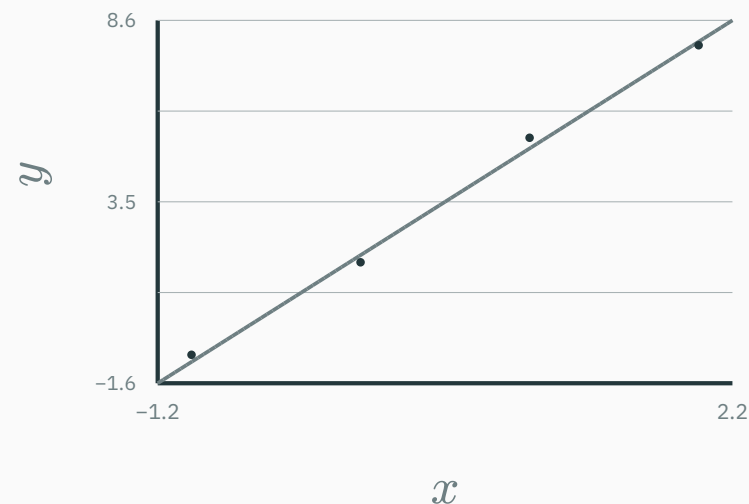
The data were generated from $f_{\text{true}(x)} = 2 + 3x$ with small measurement noise.

x	$f_{\text{true}(x)}$	observed y
-1	-1	-0.8
0	2	1.8
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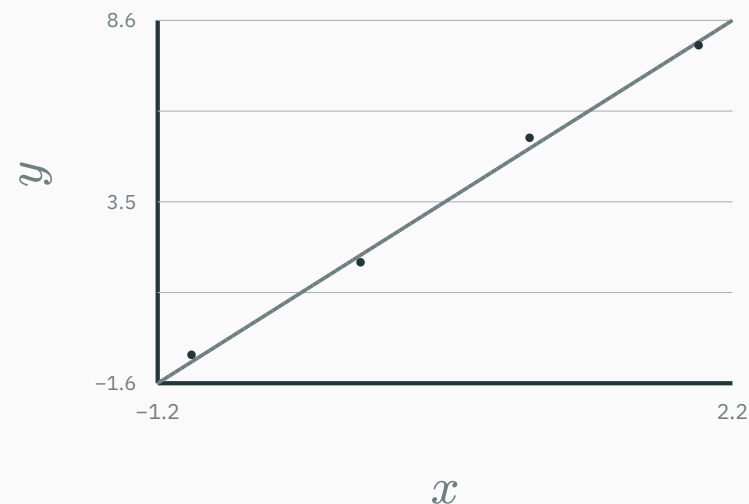


grey line = generating rule · dots = observed data

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The learner sees only the four observations and must infer the intercept b and slope w .

The likelihood peaks near the generating line

Model each observation as

$$Y_i | x_i, b, w \sim \mathcal{N}(b + wx_i, 0.5^2).$$

The data score each candidate line

$$L(b, w) \propto \exp\left(-\frac{1}{2(0.5)^2} \sum_i (y_i - b - wx_i)^2\right).$$

candidate (b, w)	squared error
(2, 3)	0.180
(0, 1)	56.580
MLE (2.07, 2.96)	0.162

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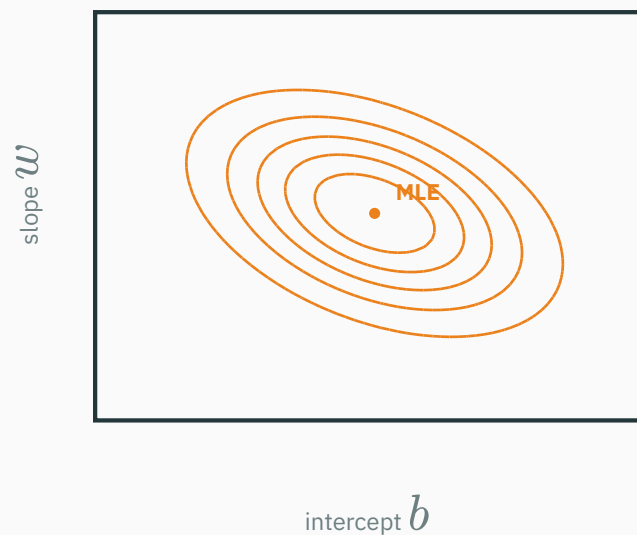
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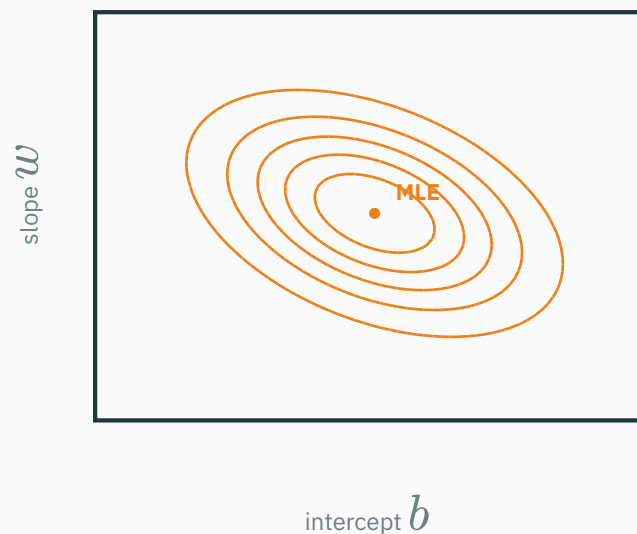
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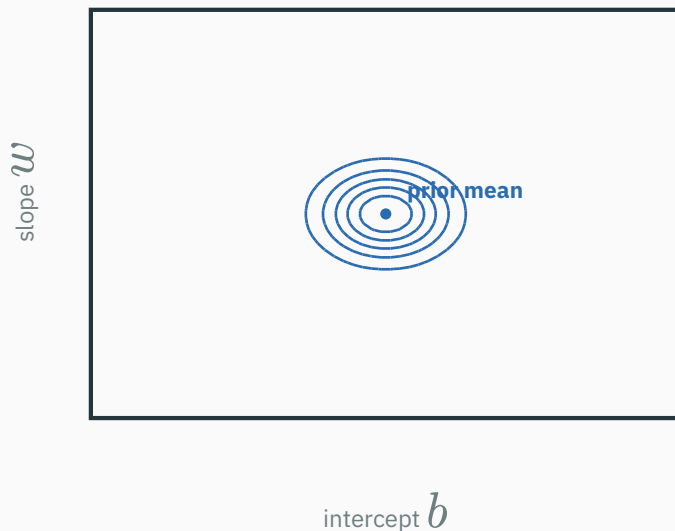
The observed data make intercepts near 2 and slopes near 3 most plausible.

Choose a simple Gaussian prior over the two regression parameters:

$$\theta = (b, w) \sim \mathcal{N}(\mathbf{0}, 0.5^2 \mathbf{I}).$$

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What does it express?

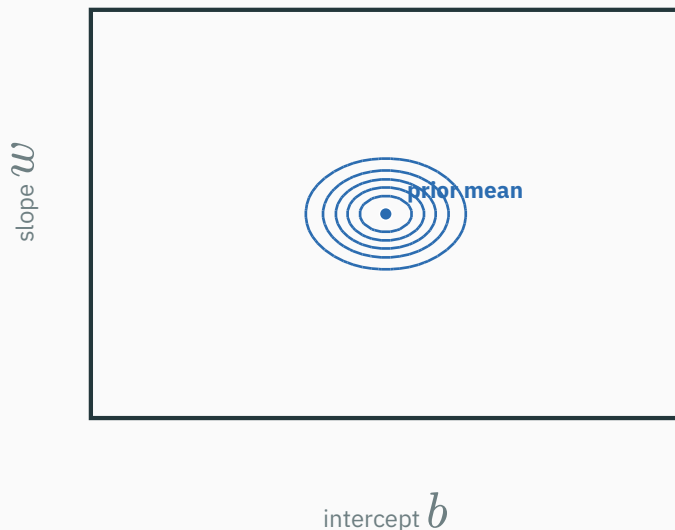
$$-\log p(b, w) = \frac{1}{2(0.5)^2} (b^2 + w^2) + C = 2(b^2 + w^2) + C.$$

Before seeing this dataset:

- $b \approx 0$: intercept near zero
- $w \approx 0$: a flatter line
- farther from $(0, 0)$: less prior density

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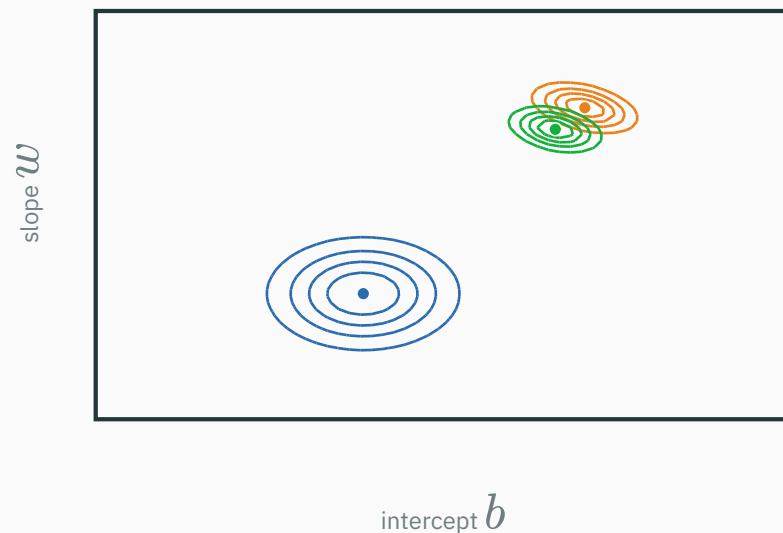
The likelihood says “fit these data”; the prior says “prefer smaller coefficients”.

Bayes combines the regression likelihood and prior

$$p(b, w \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid b, w) p(b, w)}{p(\mathcal{D})}$$

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Contours: — prior — likelihood — posterior

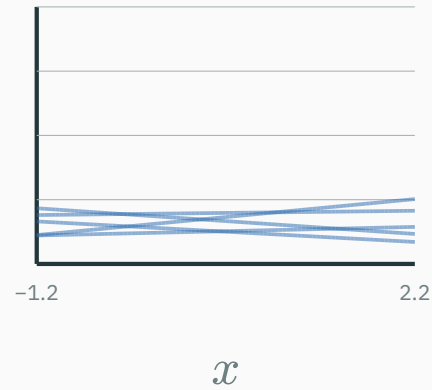
● MLE (2.07, 2.96) ● posterior peak (1.79, 2.62) ● prior mean (0, 0)

The narrower prior visibly pulls the posterior peak away from the MLE and toward zero.

The prior changes the fitted line

Prior before data

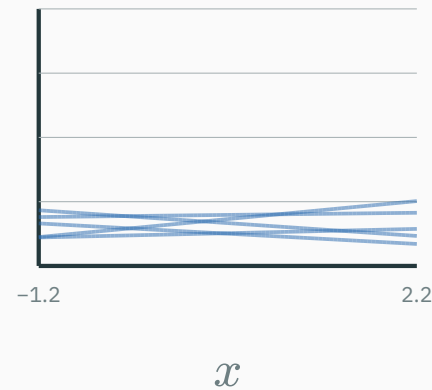
possible lines; not fitted



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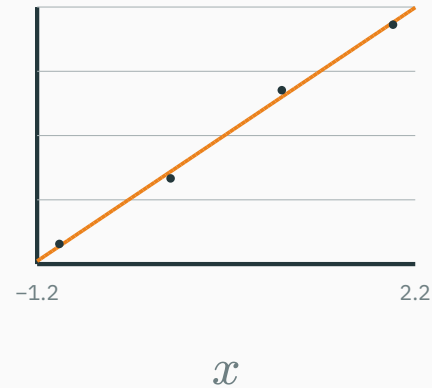
Prior before data

possible lines; not fitted



MLE

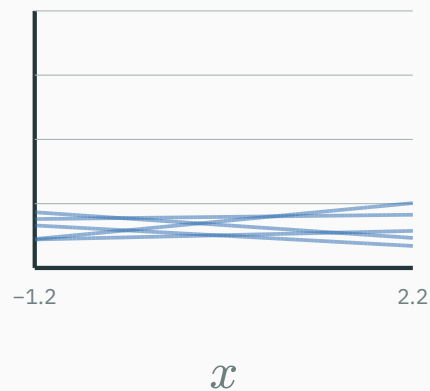
$$\hat{y} = 2.07 + 2.96x$$



The prior changes the fitted line

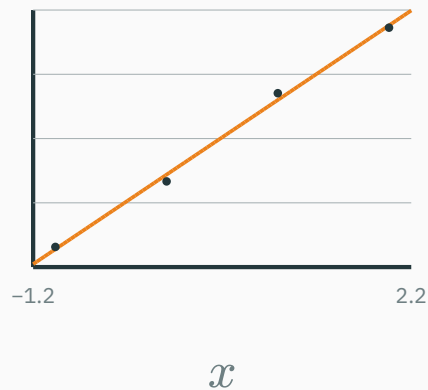
Prior before data

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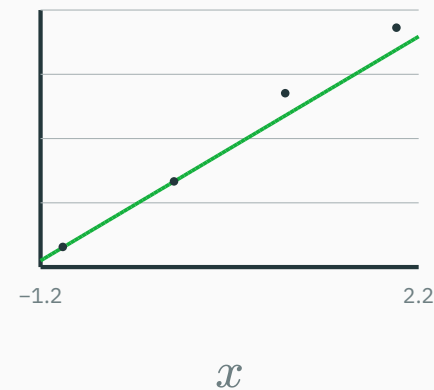
MLE

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MAP

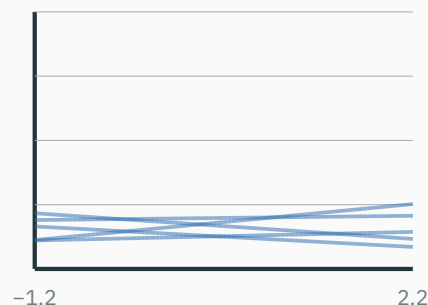
$$\hat{y} = 1.79 + 2.62x$$



The prior changes the fitted line

Prior before data

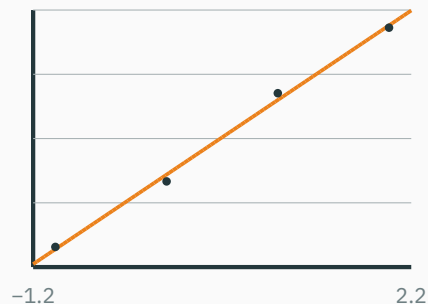
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x

MLE

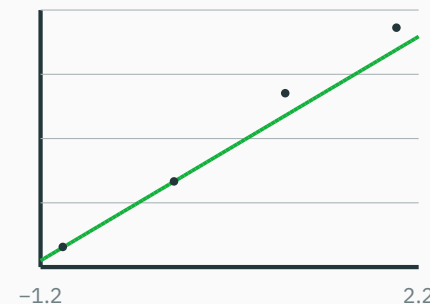
$$\hat{y} = 2.07 + 2.96x$$



x

MAP

$$\hat{y} = 1.79 + 2.62x$$



x

MLE follows these observations most closely; MAP accepts a little more error to honour the prior preference for smaller coefficients.

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} p(\theta \mid \mathcal{D}) = \arg \max_{\theta} p(\mathcal{D} \mid \theta) p(\theta)$$

$$\hat{\boldsymbol{\theta}}_{\text{MAP}} = \arg \max_{\boldsymbol{\theta}} p(\boldsymbol{\theta} \mid \mathcal{D}) = \arg \max_{\boldsymbol{\theta}} p(\mathcal{D} \mid \boldsymbol{\theta}) p(\boldsymbol{\theta})$$

Take negative logs:

$$\hat{\boldsymbol{\theta}}_{\text{MAP}} = \arg \min_{\boldsymbol{\theta}} \left(\underbrace{-\log p(\mathcal{D} \mid \boldsymbol{\theta})}_{\text{NLL}} + \underbrace{-\log p(\boldsymbol{\theta})}_{\text{negative log-prior}} \right)$$

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MAP objective: NLL + negative log-prior

In ML language: data loss + regularization.

A Gaussian prior is L_2 regularization

Assume the parameters are centred at zero before seeing the data:

$$\boldsymbol{\theta} \sim \mathcal{N}(\mathbf{0}, \tau^2 \mathbf{I}).$$

$$\underbrace{-\log p(\mathcal{D} \mid \boldsymbol{\theta})}_{\text{NLL}} + \underbrace{-\log p(\boldsymbol{\theta})}_{\text{negative log-prior}}$$

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$$-\log p(\boldsymbol{\theta}) = \frac{1}{2\tau^2} \|\boldsymbol{\theta}\|_2^2 + C$$

A Gaussian prior is L_2 regularization

Assume the parameters are centred at zero before seeing the data:

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$$\Rightarrow \mathcal{L}_{\text{MAP}} = \text{NLL} + \underbrace{\frac{1}{2\tau^2}}_{\lambda} \|\boldsymbol{\theta}\|_2^2 + C$$

$\lambda = \frac{1}{2\tau^2}$: **smaller** $\tau \rightarrow$ **larger** $\lambda \rightarrow$ **stronger shrinkage toward 0.**

Same likelihood: $L(\theta) \propto \mathcal{N}(\theta; \hat{\theta}_{\text{MLE}} = 2, s^2)$ with $s = 0.6$; prior: $p(\theta) = \mathcal{N}(0, \tau^2)$.

$$\hat{\theta}_{\text{MAP}} = \frac{\tau^2}{\tau^2 + s^2} \hat{\theta}_{\text{MLE}} \quad (\text{a shrinkage factor between 0 and 1})$$

— prior — likelihood — posterior

Prior width controls how far MAP moves

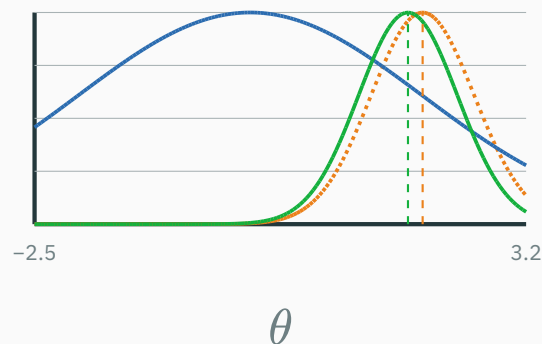
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Broad prior: $\tau = 2, \lambda = 0.125$

MLE = 2 → **MAP ≈ 1.83**



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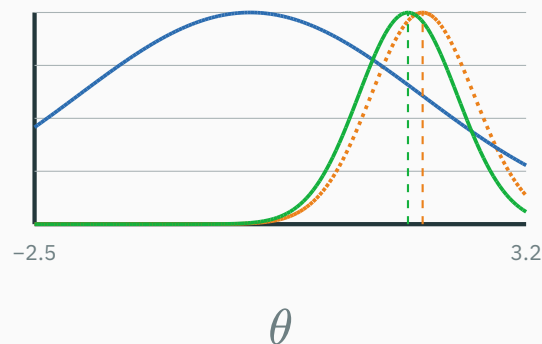
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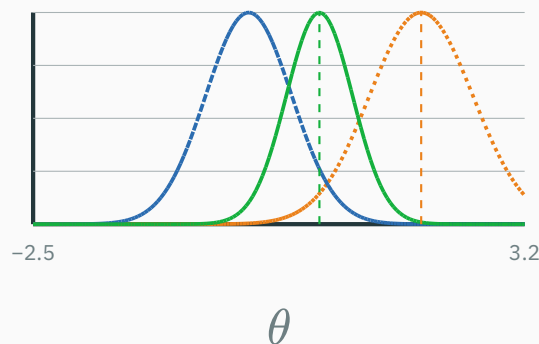
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Narrow prior: $\tau = 0.5, \lambda = 2$

MLE = 2 → **MAP ≈ 0.82**



Prior width controls how far MAP moves

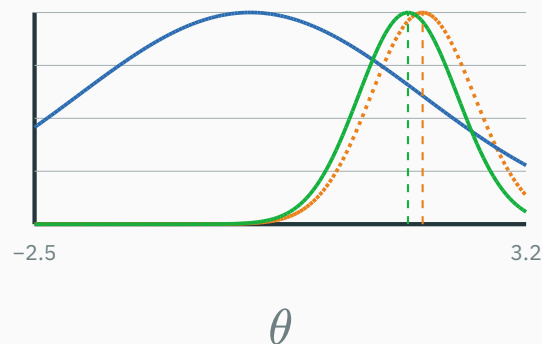
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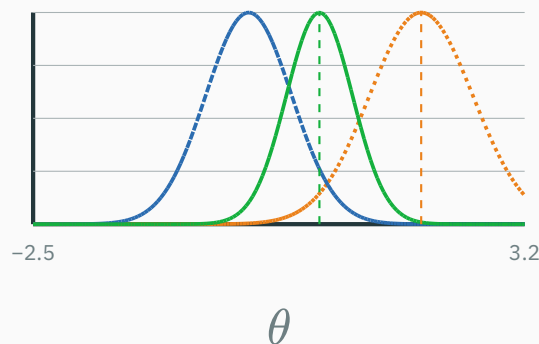
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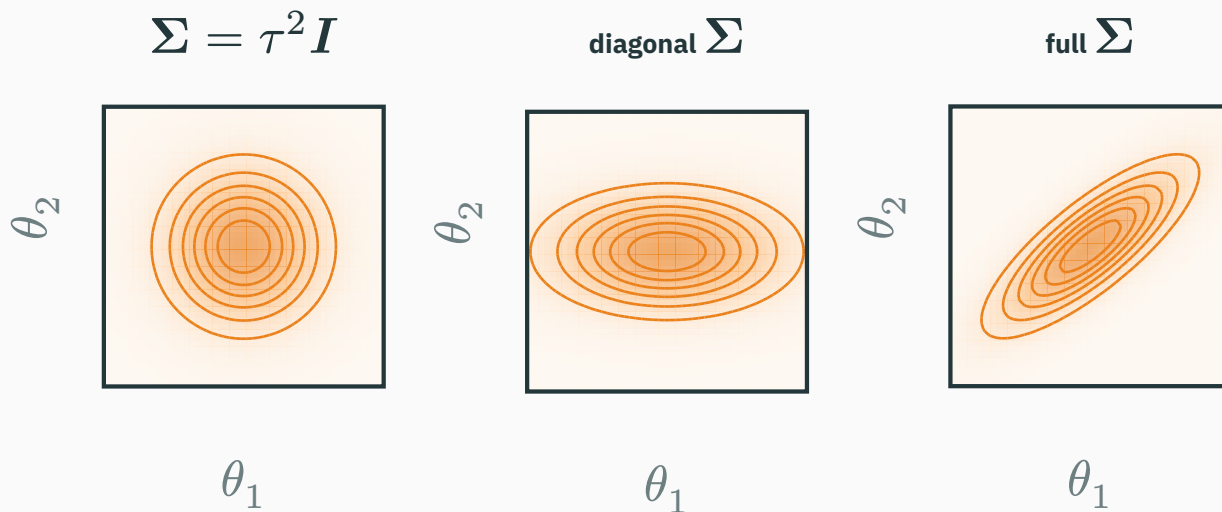


The narrower prior gives the larger λ and pulls MAP farther from the MLE toward zero.

$$\boldsymbol{\theta} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma}) \quad -\log p(\boldsymbol{\theta}) = \frac{1}{2}(\boldsymbol{\theta} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{\theta} - \boldsymbol{\mu}) + C$$

A multivariate Gaussian prior has geometry

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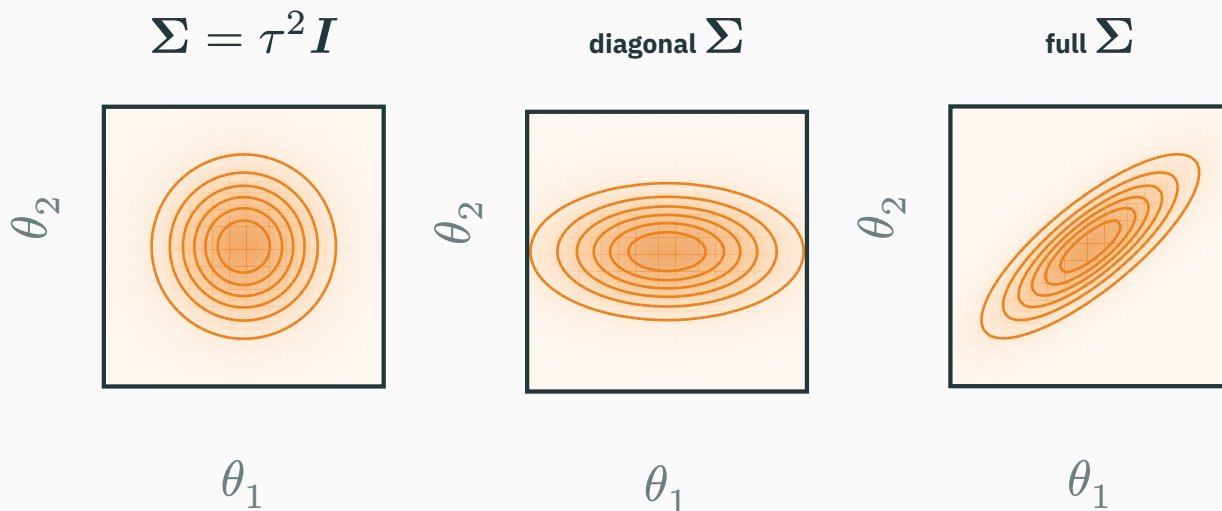


Axes: θ_1 and θ_2 are two coordinates of the parameter vector θ .

Colour = prior density $p(\theta)$: pale outer regions are less plausible; darker centres are more plausible.

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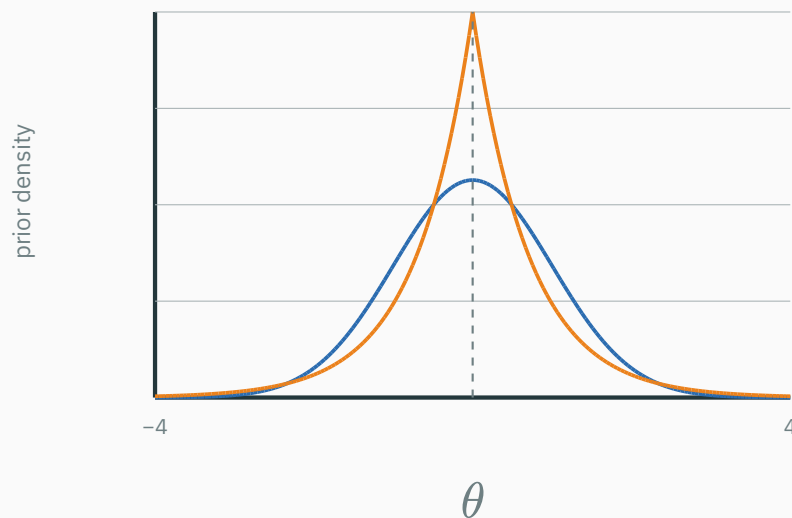
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Covariance says **which directions in parameter space are plausible.**

Laplace is sharper at zero than a Normal prior

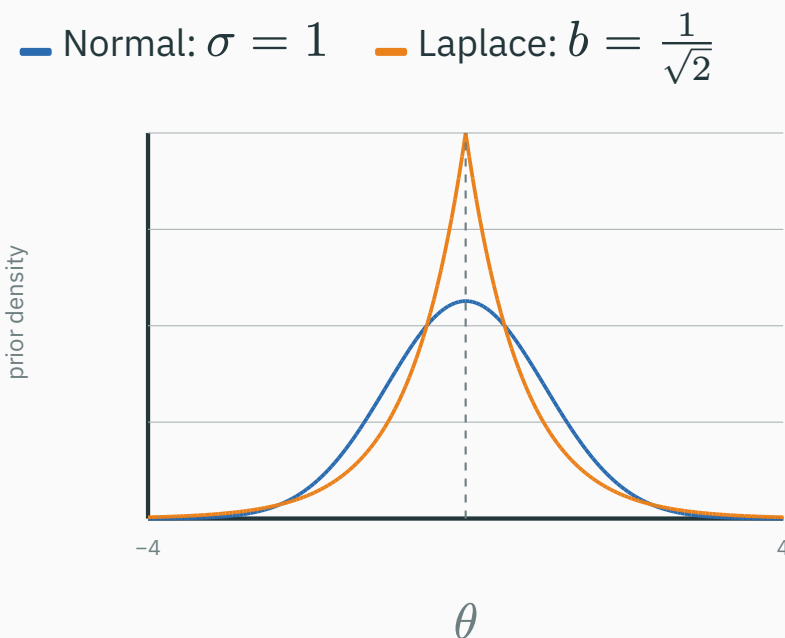
Compare zero-centred priors with the **same variance**, 1:

— Normal: $\sigma = 1$ — Laplace: $b = \frac{1}{\sqrt{2}}$



Laplace is sharper at zero than a Normal prior

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Normal prior

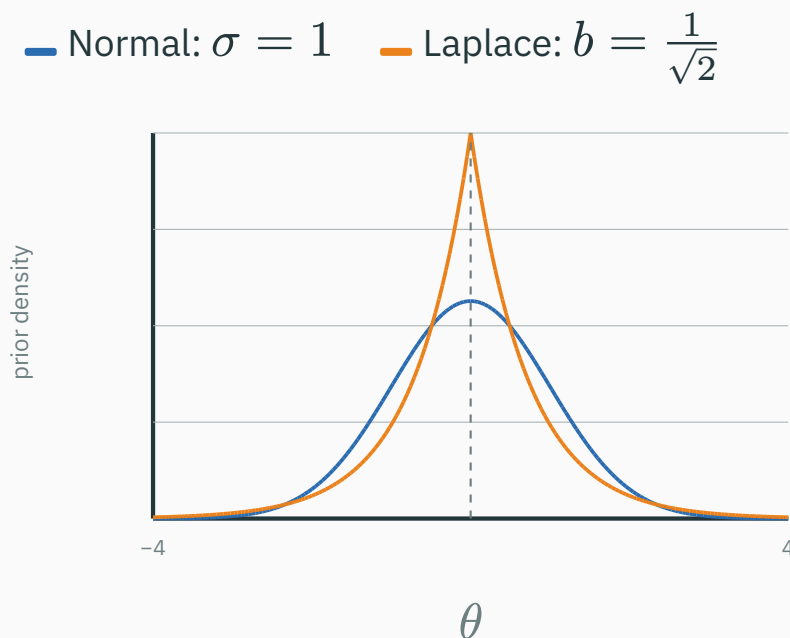
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- lighter tails

Laplace prior

- a sharp peak at $\theta = 0$
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The Laplace peak creates a **corner** in the negative log-prior—the source of L_1 sparsity.

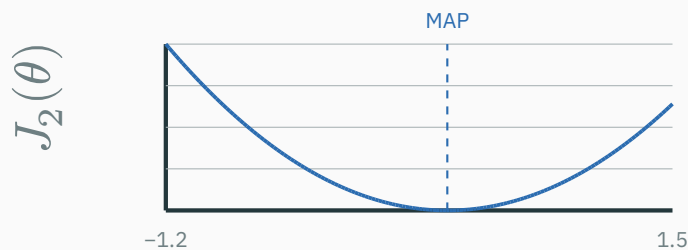
The L_1 corner can set a coefficient exactly to zero

For one coefficient, let the data term be $D(\theta) = \frac{1}{2}(\theta - 0.6)^2$, centred at $z = 0.6$.

Normal prior $\rightarrow L_2$

$$J_2(\theta) = D(\theta) + \frac{1}{2}\theta^2.$$

$$\hat{\theta}_{\text{MAP}} = \frac{0.6}{1+1} = 0.3.$$



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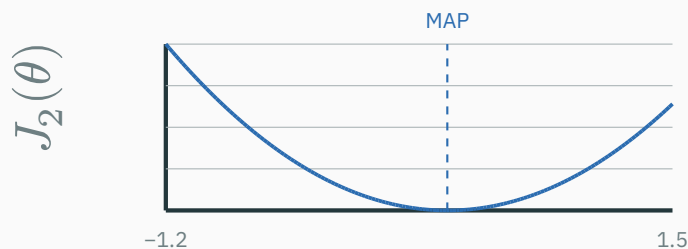
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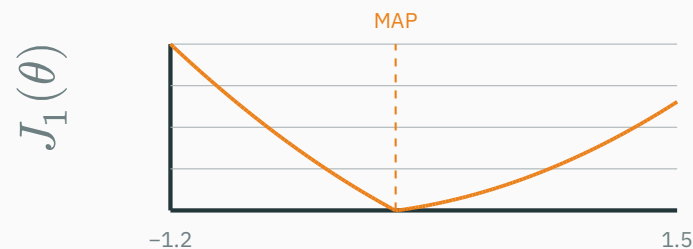
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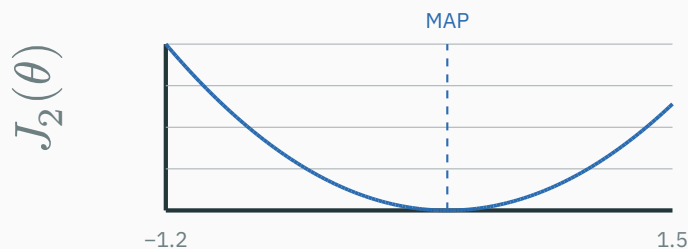
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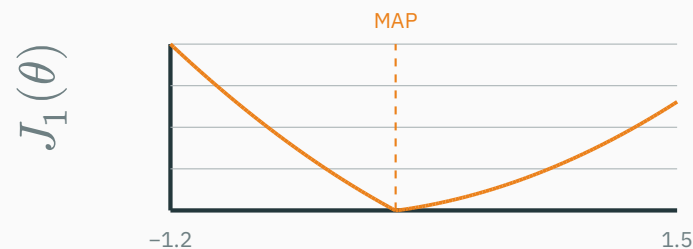
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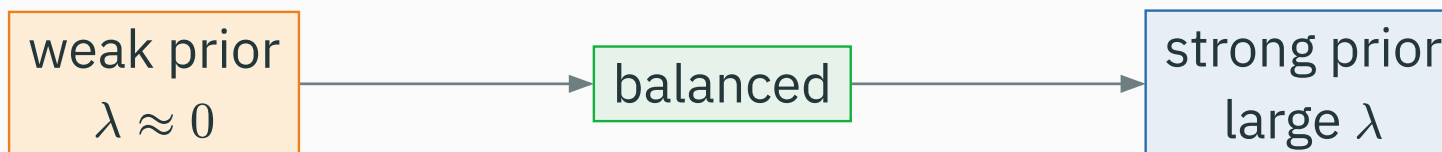
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Applied coefficient by coefficient, L_1 can make many MAP estimates **exactly zero: sparsity.**

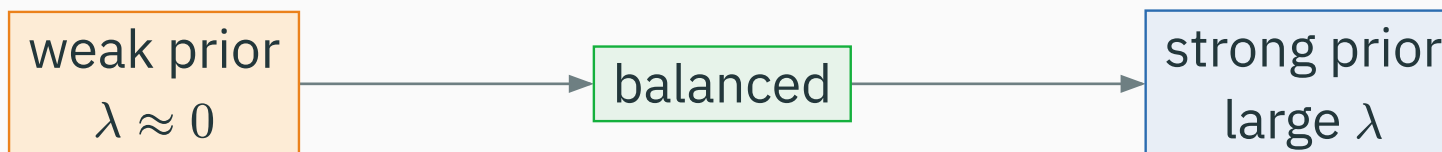
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DATA FIT DOMINATES

LIKELIHOOD + PRIOR

PRIOR DOMINATES

flexible model

high variance · overfit

enough flexibility

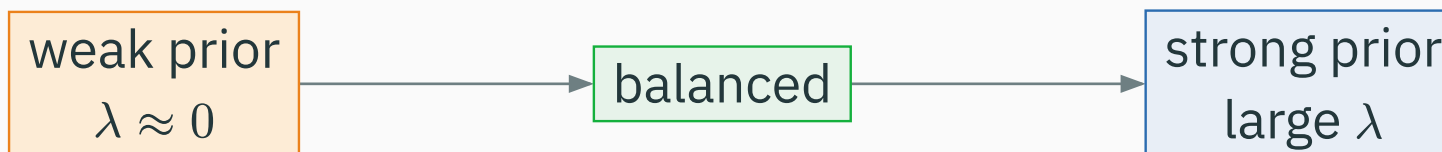
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Regularization trades variance for bias; more is not always better.

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/map-regularization

map-regularization.html — a two-parameter linear model so contours are visible. Shows **likelihood**, **prior**, and **posterior** contours with the **MLE**, **MAP**, and zero points.

Controls: amount of data · noise · prior variance · Gaussian vs Laplace · prior correlation.

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Ask: more data? narrower prior? why does MAP → MLE as data grows? a correlated prior?

Full Bayes and deep learning

MAP is still one parameter vector

MLE $\hat{\theta}_{\text{MLE}}$ — one point

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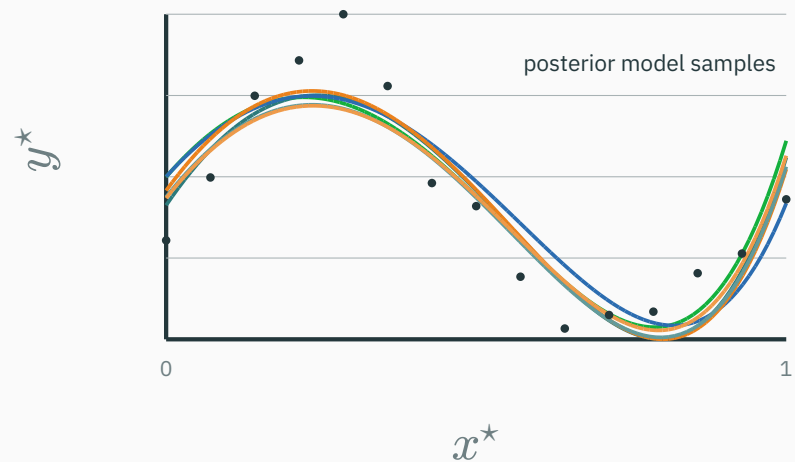
Full Bayes keeps the whole posterior $p(\theta \mid \mathcal{D})$ — a **distribution**, not a point.

Posterior predictive distribution

$$p(y^* \mid \boldsymbol{x}^*, \mathcal{D}) = \int p_{\boldsymbol{\theta}}(y^* \mid \boldsymbol{x}^*) p(\boldsymbol{\theta} \mid \mathcal{D}) \mathrm{d}\boldsymbol{\theta}$$

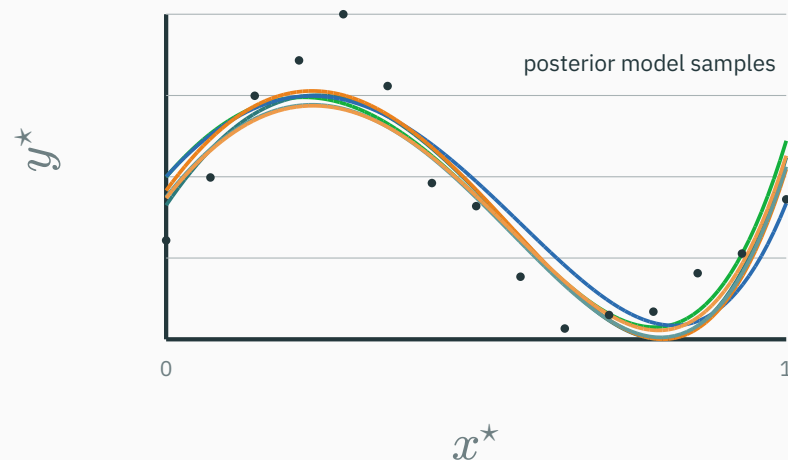
Posterior predictive distribution

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Posterior predictive distribution

$$p(y^* | x^*, \mathcal{D}) = \int p_{\theta}(y^* | x^*) p(\theta | \mathcal{D}) d\theta$$



Bayesian prediction **averages over plausible models** — and reports uncertainty.

Why DL usually uses point estimates

★ OPTIONAL

A modern network has millions to billions of parameters, so integrating $p(\boldsymbol{\theta} \mid \mathcal{D})$ is hard.

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- **variational** inference

The standard supervised DL objective

$$\min_{\boldsymbol{\theta}} \underbrace{\frac{1}{|B|} \sum_{i \in B} -\log p_{\boldsymbol{\theta}}(y_i \mid \mathbf{x}_i)}_{\text{likelihood / data fit}} + \underbrace{\lambda \|\boldsymbol{\theta}\|_2^2}_{\text{prior / regularization}}$$

With an averaged NLL, λ absorbs the dataset or minibatch scaling convention.

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probabilistic model + neural network + gradient optimization

DL doesn't discard probabilistic ML — it **scales** it with expressive functions and gradient descent.

Gaussian likelihood

⇒ **MSE**

Categorical likelihood

⇒ **cross-entropy**

Gaussian prior

⇒ L_2

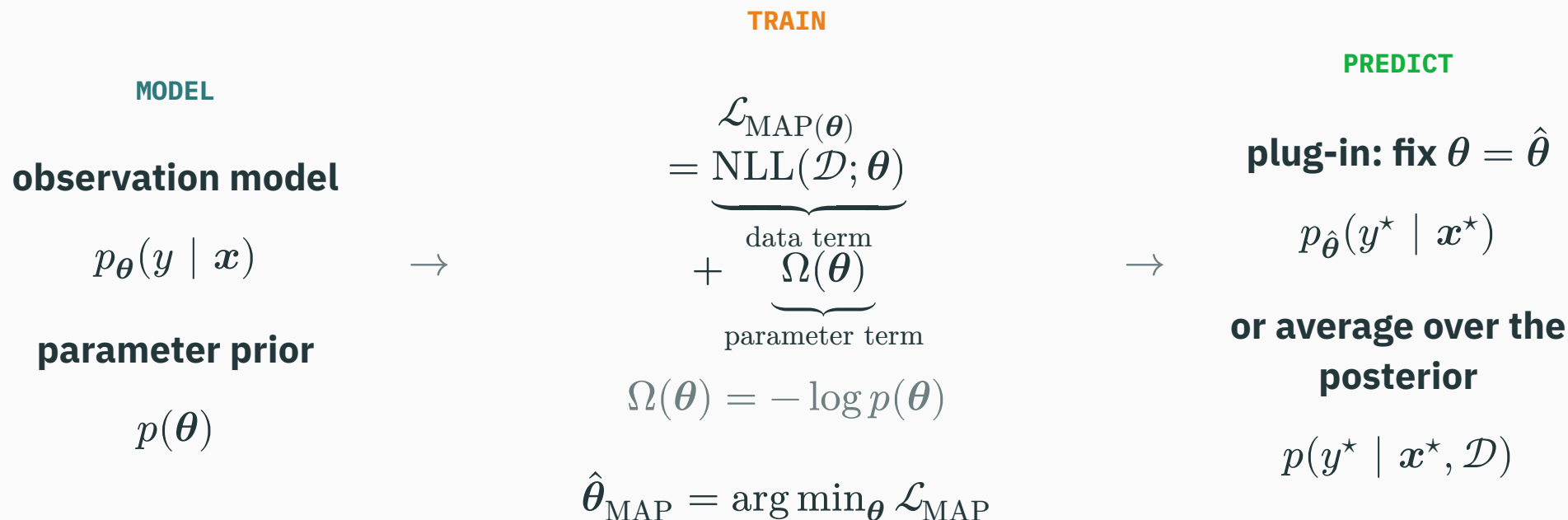
Laplace prior

⇒ L_1

MLE + prior

⇒ **MAP**

Every supervised objective begins with two assumptions



Loss $\leftrightarrow$ likelihood; regularizer $\leftrightarrow$ prior; prediction $\leftrightarrow$ plug-in or posterior predictive.

Next: from linear models to neural networks — **neurons, nonlinearities, and multilayer perceptrons.**